PHYSICS

1) Energy level A, B and C of a certain atom corresponding to increasing values of energy i.e. $E_A < E_B < E_C$. If λ_1, λ_2 and λ_3 are wavelength of radiations corresponding to transition C → B, B → A and C → A respectively, then choose CORRECT relation :-

(1) $\lambda_3 = \lambda_1 + \lambda_2$

(2) $\lambda_3 = \dfrac{\lambda_1\lambda_2}{\lambda_1 + \lambda_2}$

(3) $\lambda_1 + \lambda_2 + \lambda_3 = 0$

(4) $\lambda_3^2 = \lambda_1^2 + \lambda_2^2$

2) The total energy of a harmonic oscillator of mass 2 kg is 9 joule. If its potential energy at mean position is 5 joule, its K.E. at the mean position will be :

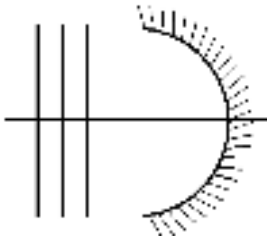
(1) 9 J

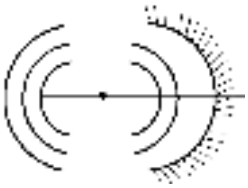
(2) 14 J

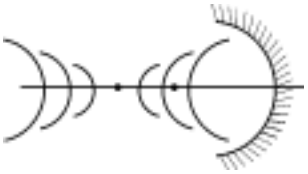
(3) 4 J

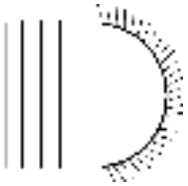
(4) 11 J

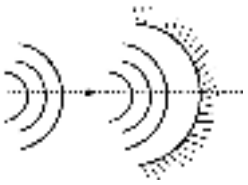
3) Plane wavefronts are incident on a spherical mirror as shown. The reflected wavefronts will be :-



(1) 

(2) 

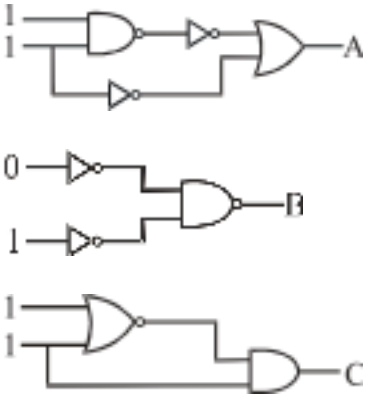
(3) 

(4) 

4) A silver ball of mass 100 gm is at a temperature 885°C. It is dropped in a silver calorimeter of mass 100 gm, filled with 170 gm of water at room temperature 30°C. Subsequently the temperature of the system is found to be 75°C. Then find specific heat of silver will be :-

(1) 0.2 cal/gm°C
(2) 0.1 cal/gm°C
(3) 0.3 cal/gm°C
(4) 0.4 cal/gm°C

5) In the following combinations of logic gates, the outputs A, B and C are respectively :-

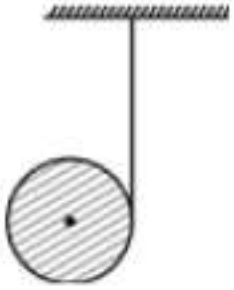


(1) 0, 1, 1
(2) 0, 1, 0
(3) 1, 1, 0
(4) 1, 0, 1

6) The ratio of speed of sound in nitrogen gas to that in helium gas at 300 K is

(1) $\sqrt{2/7}$
(2) $\sqrt{1/7}$
(3) $\sqrt{3/5}$
(4) $\sqrt{6/5}$

7) A solid cylinder of mass M and radius R starts falling under gravity as shown in figure. If mass of



chord is negligible, the acceleration of centre of mass of the cylinder will be

(1) $\frac{g}{3}$

(2) $\frac{g}{2}$

(3) $\frac{2g}{3}$

(4) g

8) The intensity of solar radiation is maximum for a wavelength 480 nm in the visible region. Estimate the surface temperature of sun: (Given Wien's constant b = 2.88×10^{-3} m K).

(1) 4000 K

(2) 6000 K

(3) 8000 K

(4) 10^6 K

9) In a Meter bridge, the null point is found at a distance of 40 cm from left end. If now a resistance of 60Ω is connected in parallel with S, connected in right gap, the null point occurs at 50 cm. Determine the value of S.

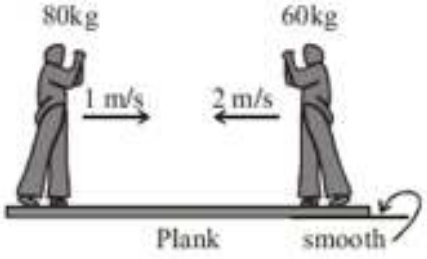
(1) $20\ \Omega$

(2) $30\ \Omega$

(3) $40\ \Omega$

(4) $70\ \Omega$

10) Two men of masses 80 kg and 60 kg are standing on a wood plank of mass 100 kg, that has been placed over a smooth surface. If both the men start moving toward each other with speeds 1 m/s and 2 m/s respectively with respect to plank then, find the velocity of the plank by which it starts moving



:

(1) $\frac{1}{6}$ m/s right side

(2) $\frac{1}{3}$ m/s left side

(3) $\frac{1}{6}$ m/s left side

(4) $\frac{1}{3}$ m/s right side

11) If $P = \frac{A^3}{B^{5/2}}$ and ΔA is absolute error in A and ΔB is absolute error in B then maximum absolute error ΔP in P is :

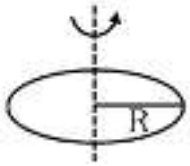
(1) $\Delta P = \pm \left(3\frac{\Delta A}{A} + \frac{5}{2}\frac{\Delta B}{B}\right) P$

(2) $\Delta P = \pm \left(3\frac{\Delta A}{A} + \frac{5}{2}\frac{\Delta B}{B}\right)$

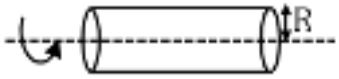
(3) $\Delta P = \pm \left(3\frac{\Delta A}{A} - \frac{5}{2}\frac{\Delta B}{B}\right) P$

(4) $\Delta P = \pm \left(3\frac{\Delta A}{B} - \frac{5}{2}\frac{\Delta B}{A}\right) P$

12) Consider a thin circular ring (A), a circular disc (B), a hollow cylinder (C) and a solid cylinder (D) of the same radius R and of same mass.

(A) 

(B) 

(C) 

(D) 

If I_A, I_B, I_C and I_D are their moments of inertia about the axis shown, then :
Choose the **correct** answer from the options given below :

(1) $I_A = 2I_B$ and $2I_C = I_D$

(2) $2I_A = I_C$ and $I_B = 2I_D$

(3) $I_A = I_B = I_C = 2I_D$

(4) $I_A = I_C$ and $2I_B = I_D$

13) **Statement-I:** When an electromagnetic wave going through vacuum is described as $E = E_0 \sin(kx - \omega t)$, then $\frac{\omega}{k}$ is independent of the wavelength.

Statement-II: $\frac{\omega}{k}$ is speed of the wave.

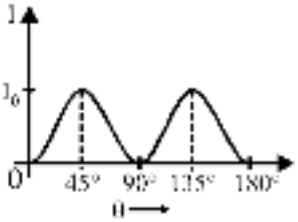
(1) Both Statement I and Statement II are correct.

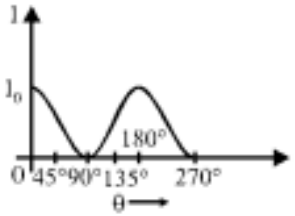
(2) Statement I is correct but Statement II is incorrect.

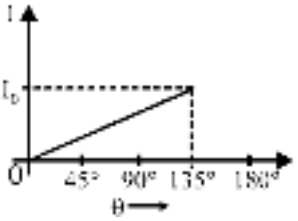
(3) Both Statement I and Statement II are incorrect.

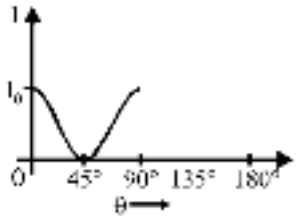
(4) Statement I is incorrect but Statement II is correct.

14) The graph showing the dependence of intensity of transmitted light on the angle between polariser and analyser is (I_0 is intensity of plane polarised light)

(1) 

(2) 

(3) 

(4) 

15) A point object is placed on the principal axis at 60 cm in front of a concave mirror of focal length 40cm. If the object is moved with a velocity of 10 cm/s along the principal axis, then the velocity (only magnitude) of image is

(1) 20 cm/s

(2) 30 cm/s

(3) 40 cm/s

(4) 60 cm/s

16) Two balls of masses m_1 and m_2 are separated from each other by a explosive powder placed between them. The whole system is at rest on the ground. Suddenly the explosive powder explodes and masses are pushed apart. The mass m_1 travels a distance s_1 and stops. If the coefficients of friction between the balls and ground are same, the mass m_2 stops after travelling the distance

(1) $s_2 = \frac{m_1}{m_2} s_1$

(2) $s_2 = \frac{m_2}{m_1} s_1$

(3) $s_2 = \frac{m_1^2}{m_2^2} s_1$

(4) $s_2 = \frac{m_2^2}{m_1^2} s_1$

17) If m, G, h, U, L, and A denote mass, graviational constant, plank's constant, Energy, angular

momentum and area respectively, then dimenions of $\frac{mh^2G}{ULA}$ is same as that of

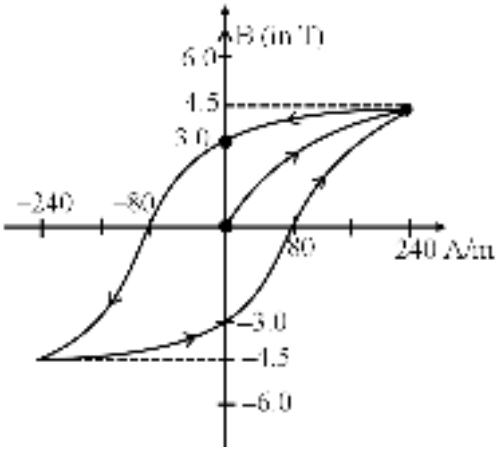
(1) Energy

(2) Power

(3) Velocity

(4) Strain

18) The figure gives experimentally measured B vs. H variation in a ferromagnetic material. The retentivity, co-ercivity and saturation, respectively, of the material are:



(1) 240 A/m, 3.0 T and 4.5 T

(2) 3.0 T, 80 A/m and 4.5 T

(3) 4.5 T, 80 A/m and 3.0 T

(4) 4.5 T, 160 A/m and 3.0 T

19) A 2 kg block is lying on a smooth table, this block is connected by a body of mass 1kg by a string which passes through a pulley. The 1kg mass is hanging vertically. The acceleration of block and tension in the string will be : ($g = 9.8\ m/s^2$)

(1) 3.27 m/s^2, 6.54 N

(2) 4.38 m/s^2, 6.54 N

(3) 3.27 m/s^2, 9.86 N

(4) 4.38 m/s^2, 9.86 N

20) An electric kettle has two heating coils. When one of the two coils is switched on, the tea begins to boil in 10 minutes. When only second coil is switched on, the boiling begins in 15 minutes. When both coils are connected in parallel and switched on the tea starts boiling in time t, then t is :-

(1) 4 minute

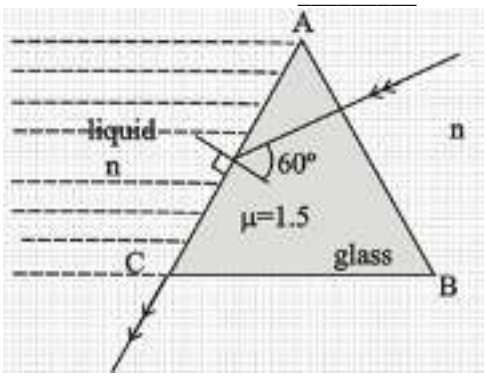
(2) 6 minute

(3) 25 minute

(4) 12 minute

21) In the given figure, the face AC of the equilateral prism is immersed in a liquid of refractive index ‘n’. For incident angle 60° at the side AC, the refracted light beam just grazes along face AC.

The refractive index of the liquid is $n = \frac{\sqrt{x}}{4}$. The value of x is_____ .



(Given refractive index of glass = 1.5)

(1) 3

(2) 9

(3) 27

(4) 36

22) Two cars C_1 and C_2 cross a point P with velocities 10 ms^{-1} and 15 ms^{-1}. After that they move with different uniform accelerations, but the car C_1 overtakes the car C_2 with a speed of 25 ms^{-1}. What is the speed of the car C_2 at that instant ?

(1) 20 ms^{-1}

(2) 25 ms^{-1}

(3) 30 ms^{-1}

(4) None of the above

23) Two identical electric point dipoles have dipole moments $\vec{p}_1 = p\hat{i}$ and $\vec{p}_2 = -p\hat{i}$ and are held on the x axis at distance 'a' from each other. When released, they move along the x-axis with the direction of their dipole moments remaining unchanged. If the mass of each dipole is 'm', their speed when they are infinitely far apart is :

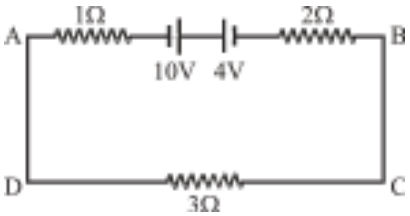
(1) $\frac{p}{a}\sqrt{\frac{1}{\pi\varepsilon_0 ma}}$

(2) $\frac{p}{a}\sqrt{\frac{3}{2\pi\varepsilon_0 ma}}$

(3) $\frac{p}{a}\sqrt{\frac{1}{2\pi\varepsilon_0 ma}}$

(4) $\frac{p}{a}\sqrt{\frac{2}{\pi\varepsilon_0 ma}}$

24) The magnitude and direction of current in following circuit will be -



(1) $\frac{7}{3}$ A from A to B

(2) $\frac{7}{3}$ A from B to A

(3) 1.0 A from B to A

(4) 1.0 A from A to B

25) The surface of water in a water tank of cross section area 750 cm^2 is h (in meter), above the tap level. The speed of water coming out through the tap of cross section area 500 mm^2 is 30 cm/s. At that instant, $\frac{dh}{dt}$ is $-X \times 10^{-3}$ m/s, where X is :

(1) 1

(2) 2

(3) 5

(4) 7

26) In an α-decay kinetic energy of α particle is 98 MeV and Q-value of the reaction is 100 MeV. The mass number of the parent nucleus is. (Assume that daughter nucleus is in ground state) :-

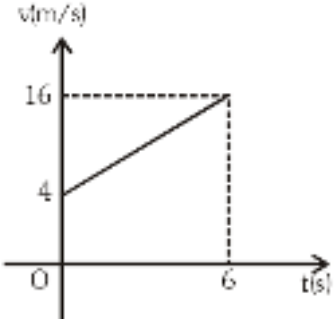
(1) 100

(2) 200

(3) 300

(4) None of these

27) Figure shows the graph of velocity versus time for a particle going along x axis. Initially at t = 0, particle is at x = –4 m. Find position of particle at t = 2s. (in m)



(1) 12 m

(2) 8 m

(3) 16 m

(4) 20 m

28) Water flows in a stream-line manner through a capillary tube of radius a. The pressure

difference being P and the rate of flow is Q. If the radius is reduced to a/2 and the pressure difference is increased to 2P, then the rate of flow becomes:-

(1) 4Q

(2) 2Q

(3) Q

(4) Q/8

29) An object viewed from a near point distance of 25 cm, using a microscopic lens with magnification '6', gives an unresolved image. A resolved image is observed at infinite distance with a total magnification double the earlier using an eyepiece along with the given lens and a tube of length 0.6 m, if the focal length of the eyepiece is equal to _____ cm.

(1) 5

(2) 15

(3) 20

(4) 25

30) Coefficient of self induction becomes $2\sqrt{2}$ times for a plane circular coil when its radius changes, keeping its number of turns same, then area of coil becomes how much times :-

(1) 8 times

(2) 4 times

(3) 2 times

(4) 16 times

31) A magnet of magnetic moment $50\,\hat{i}$ A-m^2 is placed along the x-axis in a magnetic field $\vec{B} = (0.5\hat{i} + 3.0\hat{j})$T. The torque acting on the magnet is :-

(1) $175\,\hat{k}\,N-m$

(2) $150\,\hat{k}\,N-m$

(3) $75\,\hat{k}\,N-m$

(4) $25\sqrt{37}\,\hat{k}\,N-m$

32) The thermal coefficient of expansion of a crystal in one direction (x-axis) is 2.0×10^{-6} K^{-1} and that in the other two perpendicular (y and z-axes) direction is 1.6×10^{-6} K^{-1}. What is the coefficient of cubical expansion of the crystal ?

(1) 1.6×10^{-6} K^{-1}

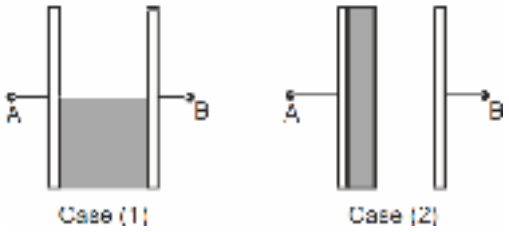
(2) 1.8×10^{-6} K^{-1}

(3) 2.0×10^{-6} K^{-1}

(4) 5.2×10^{-6} K^{-1}

33) Two identical parallel plate capacitors are filled with same dielectric material (dielectric constant K = 2) in two different ways as shown in figure. In first case the parallel plate capacitor is half filled. In both cases capacitance between A and B is same. The ratio of volume of dielectric

material between two plates in case (2) to the case (1).



(1) $\frac{5}{4}$

(2) $\frac{2}{5}$

(3) $\frac{1}{4}$

(4) $\frac{4}{3}$

34) 1 kg of water is heated from 40°C to 70°C. If its volume remains constant, then the change in internal energy is :
(Specific heat of water = 4148 J kg^{-1} K^{-1})

(1) 2.44×10^5 J

(2) 1.62×10^5 J

(3) 1.24×10^5 J

(4) 2.62×10^5 J

35)

A particle moves in a potential energy region given as $U = x^2 - 14x + 12$. Its state of equilibrium will be-

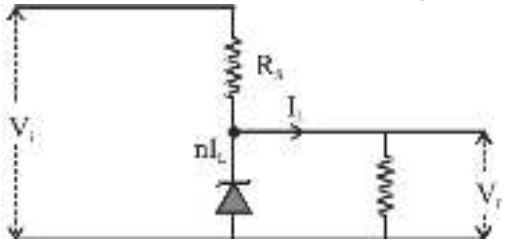
(1) x = 7 m

(2) x = 2m

(3) x = 0 m

(4) x = –1 m

36) The value of the resistor, R_S, needed in the dc voltage regulator circuit shown here, equals :-



(1) $(Vi-V_L)/nI_L$

(2) $(Vi+V_L)/n\ I_L$

(3) $(V_i+V_L)/(n+1)\ I_L$

(4) $(V_i-V_L)/(n+1)I_L$

37) A stone of mass m is tied to a string and is moved in a vertical circle of radius r making n revolutions per minute. The total tension (SI unit) in the string when the stone is at lowest point of vertical circle is

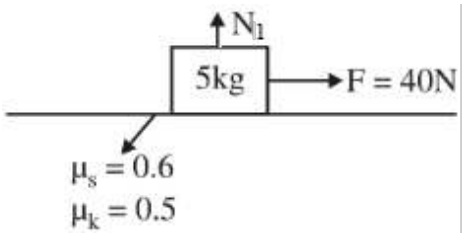
(1) mg

(2) $m(g + \pi n r^2)$

(3) $m(g + \pi n r)$

(4) $m\{g + (\pi^2 n^2 r) / 900\}$

38)



Here a is acceleration of block N_1 is normal reaction on block by ground. F_L is limiting friction, F_k is kinetic friction

Match the columns :

Column-I		Column-II	
(P)	$a(m/s^2)$	(a)	25
(Q)	N_1 (newton)	(b)	30
(R)	f_L(newton)	(c)	50
(S)	f_K (newton)	(d)	3

(1) P → c, Q → d, R → b, S → a

(2) P → d, Q → c, R → b, S → a

(3) P → a, Q → b, R → c, S → d

(4) P → d, Q → b, R → a, S → c

39) A satellite of mass m is orbiting in a circular orbit of radius $2R_e$ about the Earth. The energy required to transfer it to a circular orbit of radius $4R_e$. (Mass of earth is M_e and radius of earth is R_e).

(1) $\frac{GM_em}{2R_e}$

(2) $\frac{GM_em}{4R_e}$

(3) $\frac{GM_em}{8R_e}$

(4) $\frac{GM_em}{16R_e}$

40) An electron of mass ‘m’ and charge ‘e’ initially at rest gets accelerated by a constant electric field E. The rate of change of de-Broglie wavelength of this electron at time t; (ignoring relativistic effect) is :-

(1) $\frac{-h}{eEt^2}$

(2) $-\frac{eht}{m}$

(3) $-\frac{mh}{eEt^2}$

(4) $\frac{-h}{eE}$

41) The decreasing order of wavelength of infrared, microwave, ultraviolet and gamma rays is

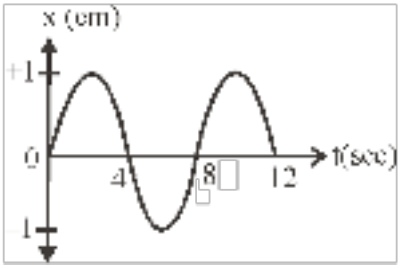
(1) microwave, infrared, ultraviolet, gamma rays

(2) infrared, microwave, ultraviolet, gamma rays

(3) gamma rays, ultraviolet, infrared, microwaves

(4) microwaves, gamma rays, infrared, ultraviolet

42) The x-t graph of a particle undergoing simple harmonic motion is shown in figure. The acceleration of the particle at t = $\frac{4}{3}$s is :-



(1) $\frac{\sqrt{3}}{32}\pi^2 cm/s^2$

(2) $\frac{-\pi^2}{32} cm/s^2$

(3) $\frac{\pi^2}{32} cm/s^2$

(4) $-\frac{\sqrt{3}}{32}\pi^2\, cm/s^2$

43) A gas expands with temperature according to the relation $V = KT^{2/3}$. Work done when the temperature of 1 mol gas is increased by 60 K is :

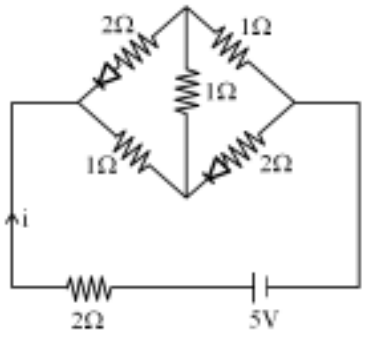
(1) 10R

(2) 30R

(3) 40R

(4) 20R

44) Current i in the network is



(1) 2.5 Amp

(2) 1 Amp

(3) $\frac{5}{7}$ Amp

(4) 0 Amp

45) n identical point charges are kept symmetrically on the periphery of the circle $x^2 + y^2 = R^2$ in xy plane. The resultant electric field at (0, 0, R) is E_1 and at (0, 0, 2R) is E_2. The ratio of $\frac{E_1}{E_2}$ is :

(1) $\frac{5\sqrt{5}}{4\sqrt{2}}$

(2) $\frac{5}{2}$

(3) $\frac{5}{4}$

(4) $\frac{5\sqrt{5}}{2\sqrt{2}}$

CHEMISTRY

1) Which should have largest difference between IE_1 and IE_2?

(1) Cl

(2) P

(3) Si

(4) Na

2) Total vapour pressure of mixture of 1 mol of volatile component A (P^o_A = 100 mm Hg) and 3 mol of volatile component B (P^o_B = 80 mm Hg) is 90 mm Hg. For such case choose correct statements:
(a) There is positive deviation from Raoult's law
(b) Boiling point has been lowered
(c) Force of attraction between A & B is smaller than that between A & A or between B & B
(d) ΔH = –ve, ΔS = +ve

(1) a, b, d

(2) a, b, c, d

(3) a, b, c

(4) b, c, d

3) 50 ml of 2M H_2SO_4 is required to neutralise 200 ml mixture of KOH and K_2SO_4. Then what is the mass of KOH present in solution ?

(1) 5.6 g

(2) 3.4 g

(3) 11.2 g

(4) 7.5 g

4) What is the volume of 1M benzoic acid solution in 300 ml of buffer solution with 2M sodium benzoate if pH of solution is 4.8 ? (pK_a = 4.20)

(1) 200 ml

(2) 300 ml

(3) 400 ml

(4) 100 ml

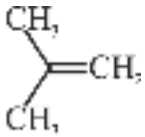
5) Which of the following compounds will give a positive Iodoform test?

(1) Propanal

(2) 2-Pentanone

(3) 3-Pentanol

(4) Benzophenone

6) P $\xrightarrow[(ii)\,H_3O^+]{(i)\,CH_3MgBr}$ Q $\xrightarrow{H^+/\Delta}$  : P is :-

(1) Acetone

(2) Acetaldehyde

(3) Tertiary butyl alcohol

(4) Isopropyl alcohol

7) **Statement I :** Generally maximum covalency of the first member of each group is four.
Statement II : Hydrogen has one unpaired electron, hence it can from maximum one covalent bond.
Select the correct option :-

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

8) Mixture of PhCHO and HCHO is treated with conc. NaOH, then cannizzaro reaction involve :-
(A) Reduction of HCHO

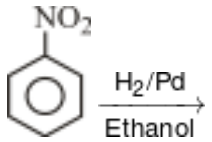
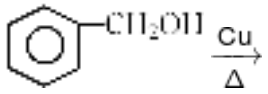
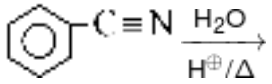
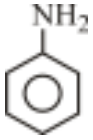
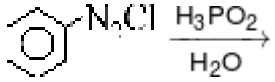
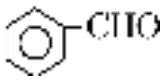
(B) Oxidation of HCHO
(C) Reduction of PhCHO
(D) Oxidation of Ph–CHO

(1) A, C
(2) A, D
(3) B, C
(4) B, D

9) Which of the following reaction of lanthanoid (Ln) is correct?

(1) $Ln + O_2 \rightarrow LnO_2$
(2) $Ln + H_2O \rightarrow LnH_3 + Ln(OH)_3$
(3) $Ln + dil.HCl \rightarrow LnCl_3 + H_2 \uparrow$
(4) $Ln + N_2 \rightarrow LnN_3$

10) Match the column

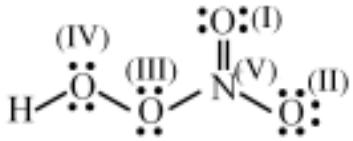
	Column-I (Reaction)		**Column-II (Product)**
(P)		(A)	
(Q)		(B)	
(R)		(C)	
(s)		(D)	

(1) P-C, Q-D, R-A,S-B
(2) P-D, Q-C, R-A,S-B
(3) P-D, Q-A, R-D,S-B
(4) P-B, Q-D, R-A,S-C

11) Which of the following ions attaches the least number of water molecules to itself when dissolved in water :-

(1) Li^+
(2) Mg^{2+}
(3) NH_4^+
(4) $[N(CH_3)_4]^+$

12) Consider the structure of HNO_4 : -



The formal charge on I, II, III, IV and V respectively are :-

(1) 0, 0, 0, 0 and 0

(2) –2, –2, –1, –1 and +5

(3) –1, +1, 0, 0 and 0

(4) 0, –1, 0, 0 and +1

13) Consider the following reactions:-

$I^- + KMnO_4 \xrightarrow{H^+} Mn^{+x} + A$

$I^- + KMnO_4 \xrightarrow{OH^-} Mn^{+y} + B$

If oxidation states of Iodine in A and B are p and q respectively, Then find the value of (x + y + p + q):

(1) 5

(2) 7

(3) 11

(4) 13

14) What is enthalpy change in following process ?

1 mol H_2O (□) → H_2O (s)

10°C –10°C

ΔH_{fusion} = 6 kJ/mol

$(C_P)H_2O$(□) = 75 J mol^{-1} K^{-1}

$(C_P)H_2O$(s) = 37.8 J mol^{-1} K^{-1}

(1) 7128 J/mol

(2) 4872 J/mol

(3) –4872 J/mol

(4) –7128 J/mol

15) The dihedral angle in staggered conformation of C_2H_6 is –

(1) 120°

(2) 0°

(3) 60°

(4) 180°

16) Which is correct order of freezing point (assuming all are strong electrolyte) ?

(1) 0.1M glucose < 0.1M NaCl < 0.2M Na_2SO_4

(2) 0.1M glucose < 0.1M Na_2SO_4 < 0.1M Na_3PO_4

(3) 0.1M glucose > 0.1M KCl > 0.1M Hg_2Cl_2 > 0.1M Na_3PO_4

(4) 0.1M glucose > 0.1M KCl > 0.1M Hg_2Cl_2 = 0.1M Na_3PO_4

17) Select correct statement about XeO_2F_2:-
(a) Bond angle : F – Xe – F > O = Xe = 0
(b) Bond length : F – Xe > O = Xe
(c) It shows folded square geometry.
(d) Number of lone pairs on central atom = 1
(e) XeO_2F_2 can be obtained by hydrolysis of XeF_6
Choose the correct answer from the option given below:-

(1) a,b,c,d only

(2) a,b,c,d,e only

(3) b,c,d only

(4) a,c only

18) Same quantity of electricity being used to liberate iodine (at anode) and a metal (at cathode). The mass of metal deposited at cathode is 0.617 g and liberated iodine completely reduces by 46.3 ml of 0.124M sodium thiosulphate solution. What is equivalent wt. of metal ? (Reaction at anode $I^{\ominus} \rightarrow I_2$)

(1) 27.7

(2) 47.4

(3) 107.47

(4) 5.7

19) Which is Aromatic species –

(1) 

(2) 

(3) 

(4) 

20) Which statement is/are correct ?
(a) Ag_3PO_4 type sparingly soluble salt have higher solubility in acidic medium.
(b) $H_2O(\square) \rightleftharpoons H_2O(g)$ at equilibrium if argon is added then amount of liquid water decreases. (constant pressure)
(c) Sum of Total no. of node in orbitals of oxygen atom (orbital containing e^-) having electron is 4.
(d) Oxidation no. of sulphur atom in $Na_2S_4O_6$ are +4 and +6.

(1) a and b only

(2) a, b, c only

(3) b, c, d only

(4) a, b, c, d

21)

Consider the reaction
$AB(g) \rightleftharpoons A(g) + B(g)$
If 33% of $AB_{(g)}$ dissociate at total pressure P atm. What is value of P if K_p = 0.4 atm :

(1) 0.4 atm

(2) 3.2 atm

(3) 6.4 atm

(4) 9 atm

22) Select correct match :-

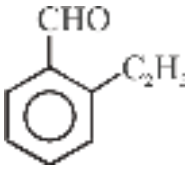
(1) $[Rh\ Br_2\ (PPh_3)_2] \rightarrow$ GI = 2 and OI = 0

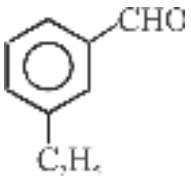
(2) $[Zn\ Br_2\ (PPh_3)_2] \rightarrow$ GI = 2 and OI = 0

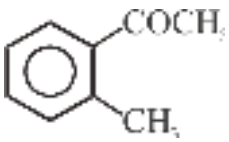
(3) $[Pt\ Br_2\ (PPh_3)_2] \rightarrow$ GI = 0 and OI = 0

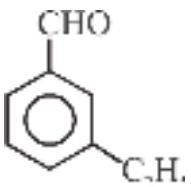
(4) $[Pt\ Cl_2\ (en)_2] \rightarrow$ GI = 2 and OI = 0

23) An aromatic compound 'X' with molecular formula $C_9H_{10}O$ gives the following chemical tests :
(i) forms 2, 4-DNP derivative
(ii) reduces Tollens' reagent
(iii) undergoes Cannizzaro reaction and
(iv) on vigorous oxidation,
1,2-benzenedicarboxylic acid is obtained.
X is :-

(1) 

(2) 

(3) 

(4) 

24) Find the incorrect statement -

(1) Carboxylic acids are reduced to primary alcohol by $LiAlH_4$.

(2) pK_a of Orthonitrophenol is lower than that of phenol.

(3) Williamson synthesis involves S_N2 attack of an alkoxide ion on primary alkyl halide.

(4) The bond between O – C of CH_3 – OH is stronger than the O–C bond of phenol

25) Number of geometrical isomers of complexes formed by mono, di, tri and tetra subsitution of Cl^- respectively in $[Pt(NH_3)_4]^{2+}$ is :-

(1) 1,2,3,4

(2) 0,2,2,2

(3) 0,2,0,0

(4) 0,2,2,0

26) $C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(\square)$, $\Delta H_{reaction}$ = –2800 kJ/mol
What is the volume of CO_2 at STP and weight of H_2O produced respectively when on combustion of certain amount of glucose 560 kJ heat is release ?

(1) 13.44 L and 21.6 g

(2) 11.2 L and 18 g

(3) 26.88 L and 12.3 g

(4) 26.88 L and 21.6 g

27) $2Fe(s) + \frac{3}{2}O_2(g) \rightarrow Fe_2O_3(s)$;
$\Delta H° = -800$ kJ/mol

$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$;
$\Delta H° = -110$ kJ/mol

What is $\Delta U°$ for following reaction at 27°C
$3C(s) + Fe_2O_3(s) \rightarrow 2Fe(s) + 3CO(g)$

(1) 470 kJ/mol

(2) 462.5 kJ/mol

(3) –470 kJ/mol

(4) –462.5 kJ/mol

28) Match the **Column I** with **Column II** :-

Column I		**Column II**	
(A)	F_2	(P)	Disproportionates with NaOH
(B)	Cl_2	(Q)	Disproportionates with water
(C)	Br_2	(R)	oxidation state changes on reaction with water

(D)	I_2	(S)	Oxidation state changes on reaction with NaOH

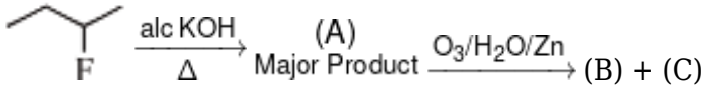
Select the correct options:-

(1) (A) QRS, (B) RS, (C) PRS (D) RS

(2) (A) QS, (B) PQR, (C) PQRS (D) PS

(3) (A) PQRS, (B) RS, (C) RS, (D) RS

(4) (A) RS, (B) PQRS, (C) PQRS (D) PS

29) 

What will be product (B) and (C) :-

(1) CH_3–CHO+CH_3–CHO

(2) CH_3CH_2CHO + HCHO

(3) 

(4) CH_3CH_2COOH + HCOOH

30) Total no. of species which can undergo disproportionation ?
S_8, $ClO_3^{\ominus}$, H_2O_2, P_4, Cl_2, Cu^+, NO_2, K^+, F_2, Na, Ag

(1) 4

(2) 5

(3) 6

(4) 7

31) Polarity of which of the following molecule changes when it undergoes lewis acid base reaction with a F^-ion?

(1) BF_3

(2) PF_5

(3) ClF_3

(4) XeF_4

32) **Assertion :** $CH_3CH_2OCH_2Cl$ reacts faster when treated with water than $CH_3CH_2OCH_2CH_2CH_2Cl$.
Reason : Carbonium ion formed by the ionisation of $CH_3CH_2OCH_2Cl$ is stabilized by resonance.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

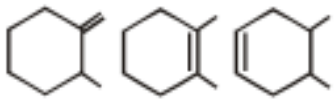
33) For first order reaction rate constant is 1×10^{-4} s^{-1} having E_a = 900 kJ/mol. Then value of lnA at T = 600 K.

(1) 50.21

(2) 171.21

(3) 430.5

(4) 349.3

34) Correct decreasing order of heat of hydrogenation is:  (I) (II) (III)

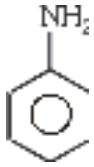
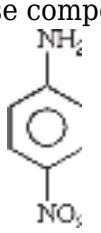
(1) I > III > II

(2) II > I > III

(3) III > I > II

(4) II > III > I

35) Arrange these compounds in decreasing order of basic strength

(i)  (ii)  (iii)  (iv) 

(1) (i) > (ii) > (iii) > (iv)

(2) (ii) > (iii) > (i) > (iv)

(3) (iv) > (i) > (iii) > (ii)

(4) (iv) > (i) > (ii) > (iii)

36)

Correct match is :

(1) ZnS - White

(2) PbS – Pink

(3) MnS – yellow

(4) CdS – Black

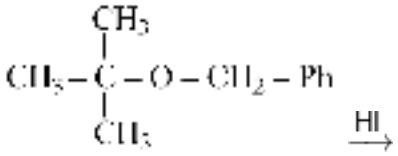
37) The RBC deficiency is deficiency disease of :

(1) Vitamin B_{12}

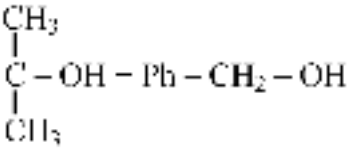
(2) Vitamin B_6

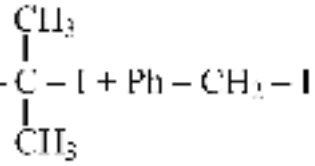
(3) Vitamin B_1

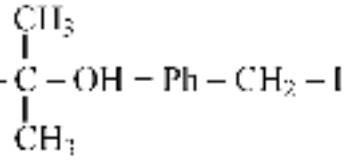
(4) Vitamin B_2

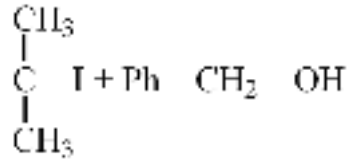
38) 

Products are –

(1) 

(2) 

(3) 

(4) 

39) $Cl^- + AgNO_3 \rightarrow A \downarrow$ (white ppt)
A + NaCN (excess)→ B
IUPAC name of B is:-

(1) Silver chloride

(2) Dicyanidosilver (I) chloride

(3) Sodium dicyanidosilver (I)

(4) Sodium dicyanidoargentate (I)

40)

The pressure of H_2 require to make the potential of H_2 electrode 0.06 V in solution having pH = 5 at 298 K.

(1) 10^{-14} bar

(2) 10^{-2} bar

(3) 10^{-12} bar

(4) 10^{-8} bar

41) If M represents the central metal atom/ion and L, a unidentate ligand than select the incorrect match:-
Complex → coordination polyhedra

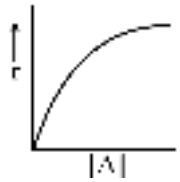
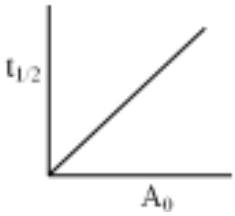
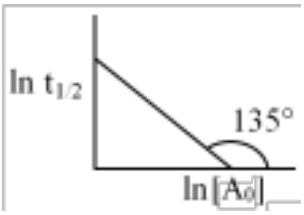
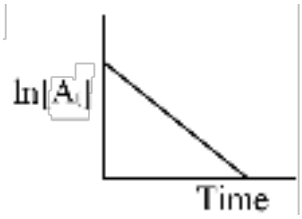
(1) ML_4 → square planar

(2) ML_5 → square pyramidal

(3) ML_3 → Trigonal bipyramidal

(4) ML_6 → square biphyramidal

42) For the reaction A → Product

	List-I (Observed rate law)		List-II (Graph)
(P)	r = k[A]	(a)	
(Q)	$r = k[A]^{1/2}$	(b)	
(R)	$r = k[A]^2$	(c)	
(S)	$r = k[A]^0$	(d)	

Select the correct match

	P	Q	R	S
(1)	d	a	c	b
(2)	b	c	a	d
(3)	a	b	c	d
(4)	d	c	b	a

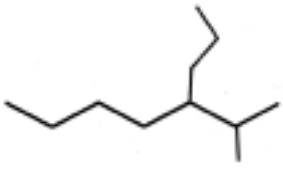
(1) 1

(2) 2

(3) 3

(4) 4

43) Give the IUPAC name for the following structure :



(1) 2-methyl-3-ethylheptane

(2) 3-ethyl-2-methylheptane

(3) 5-isopropyloctane

(4) 4-(1-methylethyl) octane

44) Which statement is/are correct ?
(A) In the titration between NaOH and CH_3COOH methyl orange is suitable indicator
(B) In the titration between $KMnO_4$ and $H_2C_2O_4$ when $KMnO_4$ is in burette then we read lower meniscus.
(C) Phenolphthalein has pink colour in unionised form (acid base titration)
(D) Oxalic acid is primary standard and $KMnO_4$ is secondary standard in titration.

(1) A and B only

(2) D only

(3) B and D only

(4) B, C, D only

45) Select the correct statement :-

(1) The dipole moment of NF_3 is more than that of NH_3.

(2) In ClO_2, all the three atoms statisfy octet rule, hence it is very stable.

(3) The bond order of C_2 and N_2^{2-} is same.

(4) The formal charge on oxygen atom in SO_4^{2-} is – 1.5.

BIOLOGY

1) Mr. Verma is suffering from hypertrichosis and phenylketonuria His father is heterozygous for phenylketonuria. The probability for Mr. Verma's sperm having one recessive autosomal allele and holandric gene is :

(1) $\frac{1}{8}$

(2) $\frac{1}{2}$

(3) $\frac{1}{4}$

(4) $\frac{1}{3}$

2) Formation of an enzyme substrate complex is :-

(1) A facultative and transient phenomenon

(2) A facultative and permanent phenomenon

(3) An obligatory and transient phenomenon

(4) An obligatory and permanent phenomenon

3) Exoskeleton of arthropods is chemically :-

(1) Cellulose

(2) Inulin

(3) Chitin

(4) Hemicellulose

4) Male and female cockroaches can be differentiate externally by :-

(1) Anal styles in male

(2) Anal style and antennae in females

(3) Anal cerci in female

(4) Both (2) and (3)

5) Read the following statements (A-D)
(A) Centrioles are absent in all plants.
(B) Cytoskeleton is made up of network of membranes.
(C) Cilia/Flagella have 9 + 2 array of axonemal microtubules.
(D) Enzymes of glyoxysome are helpful in seed germination.
Which of the above statements are wrong ?

(1) A & B

(2) B & C

(3) C & D

(4) Only B



6) Find correct option regarding this diagram:
(a) Colour of this organism appear as yellow, brown, blue and red but not green.
(b) Releases toxins that kills fishes and other marine animal
(c) Two flagella differ in their position
(d) Cell wall is made of peptidoglycan.

(1) a, b

(2) b and c

(3) a, b and c

(4) b, c and d

7) Which factor(s) is/are necessary for growth ?

(1) Water

(2) Temperature

(3) Nutrients

(4) All of the above

8) Triploblastic, metamerically segmented, bi-laterally symmetrical coelomate animals are :

(1) Platyhelminthes
(2) Annelids
(3) Aschelminthes
(4) Echinoderms

9) The renal tubule begins with a double walled cup like structure called __(A)__ which encloses the __(B)__ select the correct options regarding A & B.

(1) A - Glomerulus, B - Bowman's capsule
(2) A - Bowman's capsule, B - Glomerulus
(3) A - Bowman's capsule, B - Henley's loop
(4) A - Henley's loop, B - Glomerulus

10) **Statement-I :-** Vital capacity is the maximum volume of air a person can breath out after a forced insipiration.
Statement-II :- Inspiratory capacity is a total volume of air a person can expire after normal inspiration.

(1) Both **Statement I** and **Statement II** are correct.
(2) Both **Statement I** and **Statement II** are incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct
(4) **Statement I** is correct but **Statement II** is incorrect.

11) Which of the following is incorrectly matched ?

(1) Drupe fruit: Mango
(2) False fruit: Apple
(3) Parthenocapric fruit: Mango
(4) Legume Fruit: Pea

12) Protein encoding gene which is expressed in heterologous host produces.

(1) Foreign protein
(2) Heterologous protein
(3) Recombinant protein
(4) Alien protein

13) Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called :

(1) Plasmatic growth
(2) Maturation
(3) Development
(4) Plasticity

14) _____ play a major role in the differentation of T-lymphocyte.

(1) Melatonin
(2) Thyroxine
(3) Thymosin
(4) Thyrocalcitonin

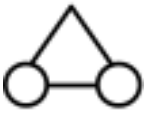
15) Gout is the inflammation of joints due to Accumulation of _____ crystals ?

(1) Folic Acid
(2) Urea
(3) Uric acid
(4) Ammonia

16) For the AB blood group system, the frequencies of A and B alleles are 0.6 and 0.4, respectively. The expected frequency of AB blood group bearing organisms is likely to be :-

(1) 24%
(2) 48%
(3) 36%
(4) 16%

17) Match the following-

	Column-I		**Column-II**
(i)		(A)	Monozygotic twins
(ii)		(B)	Carrier for sex linked recessive disease
(iii)		(C)	Carrier for autosomal recessive disease
(iv)		(D)	Normal female

(1) (i–A), (ii–B), (iii–C), (iv–D)
(2) (i–D), (ii–C), (iii–A), (iv–B)
(3) (i–C), (ii–D), (iii–B), (iv–A)
(4) (i–B), (ii–C), (iii–D), (iv–A)

18) Match List-I with List-II :-

	List-I		**List-II**

(A)	Globulins	(i)	Responsible for immune response of body
(B)	Thrombocytes	(ii)	6000-8000/mm^3 blood
(C)	Leucocyte	(iii)	Primarily involve in defense mechanism
(D)	B and T lymphocyte	(iv)	Produced from megakaryocytes of bone marrow.

(1) A-(iii), B-(ii) ; C - (i) ; D-(iv)

(2) A-(iv), B-(ii) ; C - (i) ; D-(iv)

(3) A-(iii), B-(iv) ; C - (ii) ; D-(i)

(4) A-(iii), B-(ii) ; C - (iv) ; D-(i)

19) Match **List-I** with **List-II**.

	List-I		**List-II**
(A)	Antibody	(I)	Intercellular ground substance
(B)	Receptor	(II)	Fights infectious agents
(C)	Trypsin	(III)	Sensory reception
(D)	Collagen	(IV)	Enzyme

Choose the **correct** answer from the options given below :

(1) A-IV, B-I, C-II, D-III

(2) A-I, B-II, C-III, D-IV

(3) A-II, B-III, C-IV, D-I

(4) A-III, B-IV, C-I, D-II

20) Match the column-I with column-II :

Column-I (Classes)		**Column-II (Examples)**	
(A)	Psilopsida	(I)	*Porphyra*
(B)	Rhodophyceae	(II)	*Equisetum*
(C)	Sphenopsida	(III)	*Psilotum*
(D)	Pteropsida	(IV)	*Adiantum*

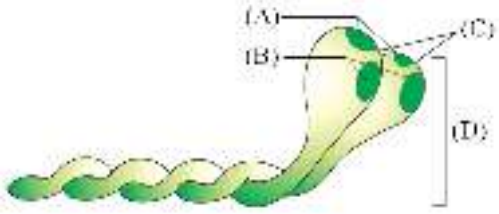
(1) (A)→III ; (B)→II ; (C)→I ; (D)→IV

(2) (A)→III ; (B)→I ; (C)→II ; (D)→IV

(3) (A)→III ; (B)→IV ; (C)→II ; (D)→I

(4) (A)→I ; (B)→II ; (C)→III ; (D)→IV

21) Identify the figure & choose the correct matching :-



(1) A → Head, B → Cross arm,
C → Actin binding site,
D → ATP binding site

(2) A → Actin binding site,
B → ATP binding site,
C → Head, D → Cross arm

(3) A → Actin binding site,
B → ATP binding site,
C → Head, D → Tail

(4) A → ATP binding site,
B → Actin binding site,
C → Head, D → Cross arm

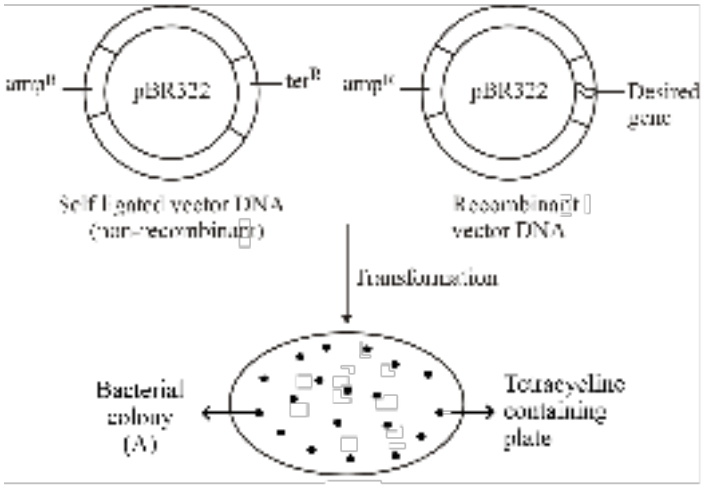
22) Select the incorrect one in reference to Wobble hypothesis.

(1) Maintains economy of tRNA

(2) Specificity of an anticodon is particularly determined by first two bases of a codon.

(3) Major cause of degeneracy is the first two N-bases of codon.

(4) Third base of a codon shows vibrating capacity.

23) 

Which is correct for 'A' :-

(1) Transformant with non-recombinant DNA

(2) Transformant with recombinant DNA

(3) Non-transformant

(4) Both (1) & (2)

24)

Match List-I with List-II :

	List-I		List-II
(A)	Ascomycetes	(I)	*Albugo*
(B)	Basidiomycetes	(II)	*Neurospora*
(C)	Deuteromycetes	(III)	*Ustilago*
(D)	Phycomycetes	(IV)	*Trichoderma*

Choose the option with all correct matches :

(1) A-I, B-II, C-IV, D-III

(2) A-II, B-III, C-IV, D-I

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-III, C-I, D-IV

25) Read the following statements :
(a) In hydra, neural system is composed of a network of neuron.
(b) Vertebrates have a less developed neural system.
(c) Neural system is better organised in insect.
(d) Neural system of animals is composed of highly specialised cells called neurons.
Choose the correct options.

(1) a, b, c

(2) a, c, d

(3) a, b, d

(4) b, d, c

26) **Assertion (A) :** Adrenaline and nor-adrenaline hormones are called emergency hormones or hormones of fight or flight.
Reason (R) : These hormones are rapidly secreted in response to emergency condition.

(1) A and R both correct, R is correct explanation of A.

(2) A and R both correct, R is not correct explanation of A.

(3) A correct, R incorrect

(4) Both A and R incorrect

27) In the larval stage of star-fish which type of symmetry is present ?

(1) Radial symmetry

(2) Asymmetry

(3) Bilateral symmetry

(4) None of Above

28) **Assertion (A) :** Almost all enzymes are proteins but some nucleic acids also behave like enzyme.
Reason (R) : Enzymes require optimum pH & temperature for maximum activity.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) Both **Assertion** and **Reason** are false.

29) Match List -I with List II

List -I		**List-II**	
A	Typhoid	i	Widal test
B	Common cold	ii	Auto immune disease
C	Rheumatoid arthritis	iii	Rhinovirus
D	Ring worm	iv	Microsporum

Choose the correct answer from the options given below.

(1) A - i, B - iv, C - iii, D - ii

(2) A - iv, B - i, C - ii, D - iii

(3) A - iv, B - i, C - iii, D - ii

(4) A - i, B - iii, C - ii, D - iv

30) Arrange the following in the correct order of sperm formation, maturation and transport in human males:
(A) Epididymis
(B) Vas deferens
(C) Seminiferous tubules
(D) Ejaculatory duct
(E) Rete testis
(F) Vasa efferentia

(1) C → E → F → A → B → D

(2) C → F → E → A → B → D

(3) E → C → F → A → D → B

(4) F → C → E → A → B → D

31) **Assertion (A) :** The large holes in 'Swiss cheese' are due to production of a large amount of CO_2 by a bacterium.
Reason (R) : The 'Roquefort Cheese' are ripened by growing a specific fungi on them, which gives them a particular flavor.

(1) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion.

(2) Both Assertion and Reason are correct and Reason is not the correct explanation for Assertion.

(3) If Assertion is true but the Reason is false.

(4) If Assertion is false but the Reason is true.

32) Association areas are present in which structure of brain?

(1) Hypothalamus

(2) Cerebral cortex

(3) Cerebellum

(4) Mid brain

33) A molecular biologist is investigating the thermal stability of a double stranded DNA. He isolated a DNA from thermophilic bacterium as 'A' and another DNA from a mesophilic bacterium labelled as 'B'. Both the DNA of same size and subjected to high temperature i.e. 80°C. After this when both A & B are separated by gel electrophoresis, three bands of DNA appeared.
Conclusion of the experiment is-

(1) Both the DNA 'A' & 'B' denatures at 80°C temperature.

(2) DNA from mesophilic bacteria 'B' denatures at 80°C but DNA from thermophilic bacteria 'A' does not.

(3) Neither DNA 'A' nor 'B' denatures at 80°C temperature.

(4) There is no effect of temperature on denaturation of DNA 'A' & 'B'.

34) Given below are two statements:
Statement-I : Connel's elegant field experiments showed that on the rocky sea coast of Scotland, the larger and competitively superior barnacle Balanus dominantes the intertidal areal and excludes the smaller barnacle chathalamus from that zone.
Statement-II : The life cycle of parasites is often simple, involving one or two intermediate host or vectors to facilitate parasitisation of its primary host.
In the light of above statements, choose the correct answers from the options given below :

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

35) Which of the following statement is correct ?

(1) Cyanobacteria have all pigments similar to green plants.

(2) Cyanobacteria are photosynthetic autotrophs.

(3) Cyanobacteria are exclusively marine

(4) Cyanobacteria show true sexual reproduction.

36) **Assertion :** Notochord is a mesodermally derived rod like structure present on the dorsal side of body.
Reason : Animal which possess notochord are called chordate.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

37) **Statement-I :** Regulation of lac operon by repressor is referred to as negative regulation.

Statement-II : HGP was closely associated with the rapid development of a new area in biology called Bioinformatics.

(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct and Statement-II are incorrect.
(3) Statement-I is incorrect and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

38) During ventricular systole :-

(1) Tricuspid and bicuspid values are open.
(2) Semilunar valves are open.
(3) All the four chambers of heart are in a relaxed state.
(4) Ventricular pressure is low

39) **Statement-I :** An organism represents/occupies a place or position in the system of classification.
Statement-II : There are many categories/ranks in classification which are generally referred to as taxonomic categories or taxa.

(1) **Statement I** is correct but **Statement II** is incorrect.
(2) **Statement I** is incorrect but **Statement II** is correct
(3) Both **Statement I** and **Statement II** are incorrect.
(4) Both **Statement I** and **Statement II** are correct.

40) Most common respiratory substrate are :-

(1) Organic acids
(2) Fats
(3) Proteins
(4) Carbohydrates

41) Which one of the following statements are correct regarding ecosystem?
(A) Net Primary Productivity (NPP) of an ecosystem represents the biomass available for herbivores and decomposers.
(B) The rate of decomposition is faster when detritus is rich in lignin and chitin, but slower when it contains nitrogen and water-soluble substances.
(C) In any ecosystem, the pyramid of energy is always upright, while the pyramid of biomass in the sea may be inverted.
(D) A grazing food chain begins with green plants, whereas a detritus food chain begins with dead organic matter.
(E) Forest, grassland and desert are examples of terrestrial ecosystems, while pond, lake wetland and estuary are aquatic ecosystems.

(1) Only A, C and D are correct.
(2) Only A, B and E are correct.
(3) Only A, C, D and E are correct.

(4) All statements are correct.

42) **Assertion :** Metacentric chromosome has middle centromere forming two almost equal arms of the chromatid.
Reason : Duration of cell cycle can vary from organism to organism and also from cell type to cell type.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

43) Which of the following is favorable for left shift of oxygen dissociation curve :-

(1) High Po_2 and high Pco_2
(2) High Po_2 and lesser H^+ concentration
(3) Low Pco_2 and high H^+ concentration
(4) Low Pco_2 and high temperature

44) Which type of muscles are present in the wall of blood vessels, stomach and intestine ?

(1) Cardiac muscles
(2) Involuntary and nonstriated muscles.
(3) skeletal muscles
(4) Striped muscles.

45) Match the columns and choose correct option ?

	Column I		**Column II**
a	Recombinant technologies	i	Large volumes (100-1000 litres) of culture can be processed
b	Continuous culture system	ii	The ultimate aim is to produce a desirable protein
c	Bioreactors	iii	Used to produces a larger biomass leading to higher yields of desired products

(1) a-i, b-ii, c-iii
(2) a-ii, b-iii, c-i
(3) a-iii, b-i, c-ii
(4) a-ii, b-i, c-iii

46) Which of the following is not an example of perigynous flower ?

(1) Peach
(2) Plum

(3) Pea
(4) Rose

47) Which of the following are pseudocoelomate –

(1) Arthropods
(2) Mollusca
(3) Platyhelminthes
(4) Aschelminthes

48) Which of the following restriction site/sites are present at tet^R gene in pBR322 vector.

(1) Bam HI
(2) Sal I
(3) Pvu I
(4) Both 1 & 2

49) Which of the following statement is not correct for slime moulds ?

(1) They are saprophytes
(2) During unfavourable conditions they form plasmodium
(3) Their spores are dispersed by air currents
(4) Their spores have true walls

50) Mitotic divisions of the zygote named cleavage starts when :-

(1) Zygote reaches into uterine cavity.
(2) The zygote moves through the infundibulum of the oviduct.
(3) Implantation inside uterine endometrium has take place.
(4) The zygote moves through the isthmus of the oviduct.

51) Transfer of pollen grains from anther to the stigma of a different plant is :-

(1) Autogamy
(2) Geitonogamy
(3) Xenogamy
(4) Cleistogamy

52) Which of the following statements is **incorrect** ?

(1) leaf pigments of any green plant can be separated through paper chromatography.
(2) Chlorophyll 'a' is visualised as bright/blue green colour on chromatogram.
(3) The maximum absorption by Chlorophyll 'a' is found in the blue and red regions
(4) Chlorophyll 'b' is the chief pigment associated with photosynthesis.

53) World summit on sustainable development was held in 2002 in :-

(1) Rio de Janeirio
(2) Montreal
(3) Kyoto
(4) Johannesburg

54) The menstrual flow result due to :-

(1) Rupturing of graafian follicle
(2) Rupturing of ovarian wall
(3) disintegration of uterine tissue
(4) All of the above

55) Which of the following is correct for the position of xylem in leaves ?

(1) Above the vascular cambium
(2) Below the vascular cambium
(3) Towards the abaxial surface
(4) Towards the adaxial surface

56) 'ABO' blood group is not an example of :-
(A) Co dominance
(B) Multiple allele
(C) Incomplete dominance
(D) Pleiotropic gene
(E) Polygenic inheritance

(1) A, B, C, D, E
(2) C, D, E
(3) A, B
(4) A, B, C

57) Out of following which statement is correct for nervous system of frog ?

(1) Forebrain includes olfactory lobes, unpaired cerebral hemisphere and paired diencephalon.
(2) The mid brain is characterised by two pair of optic lobes.
(3) Hind brain consists of cerebellum and medulla oblongata.
(4) There are 12 pairs of cranial nerves arising from brain

58) Which complex related to oxidative phosphorylation has two copper centers ?

(1) Complex II
(2) Complex IV
(3) Complex V
(4) Complex III

59) Read the following statements :

(i) Promotes female flowers in cucumber.
(ii) Initiate germination in peanut seeds.
(iii) It is used to speed up the malting process.
(iv) It helps in synchronising fruit set in Pineapples.
Which statements from the above are **true** about Ethylene ?

(1) (i), (iii), (iv)

(2) (i), (ii), (iii)

(3) (i), (ii), (iv)

(4) (i), (ii), (iii), (iv)

60) Which of the following statement is **incorrect** regarding reproductive fitness of an organism?

(1) Bamboo breed only once in their lifetime.

(2) Pelagic fishes produce a large number of large sized offspring.

(3) Birds produce a small number of large-sized offsupring.

(4) All of the above

61) Which of the following statement are true.
(a) Tilman found that plots with more species showed more year to year variation in total biomass.
(b) The IUCN Red list documents the extinction of 784 species in the last 500 years.
(c) The last twenty years alone have witnessed the disappearance of 37 species.
(d) In general loss of biodiversity in a region may lead to decline in plant production.

(1) a and c are correct

(2) a and b are correct

(3) b and d are correct

(4) b and c are correct

62) In animal cells, during the S phase, DNA replication begins in the ..A.. and the centriole duplicates in the ..B.. .
Here A and B are :-

(1) A - Nucleus B - Nucleus

(2) A - Cytoplasm B - Cytoplasm

(3) A - Nucleus B - Cytoplasm

(4) A - Cytoplasm B - Nucleus

63)

Which of the following is true regarding the given diagram ?



(1) Chlorophyll a, d and carotenoids are present

(2) Male gametes are motile

(3) Zoospores are thin wall and have two flagella

(4) 1 and 2 both

64)

Match the column-I with column-II :-

	Column-I		**Column-II**
(A)	Vault	(I)	Hormone releasing IUD's
(B)	Saheli	(II)	Oral pill
(C)	Lippes loop	(III)	Barrier method
(D)	LNG 20	(IV)	Non-medicated IUD's

(1) A-III, B-II, C-I, D-IV

(2) A-IV, B-II, C-III, D-I

(3) A-IV, B-III, C-I, D-III

(4) A-III, B-II, C-IV, D-I

65) **Statement-1 :** Main plant body of pteridophytes is sporophyte.
Statement-2 : Sporangia in pteridophytes are borne on leaf like appendages called sporophylls.

(1) Both statements are correct.

(2) Both statements are incorrect.

(3) Statement-1 is correct, statement-2 is incorrect.

(4) Statement-1 is incorrect, statement-2 is correct.

66) When ready made antibodies are directly given to protect the body against foreign agents, it is called ____-

(1) Innate immunity

(2) Passive immunity

(3) Active immunity

(4) Both (1) and (3)

67) Which one of the following hormone is not secreted by gastro intestinal tract :

(1) Gastrin

(2) Secretin

(3) CCK

(4) ANF

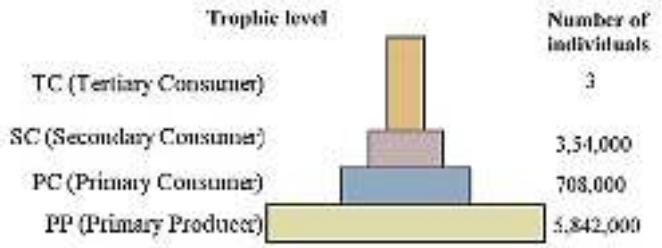
68) Read the statement carefully :
(1) Morphine is very effective sedative and painkiller.
(2) These days cannabinoids are being abused by some sports persons.
(3) Natural cannabinoids are obtained from the inflorescences of the plant papaver somniferum.
(4) Coca alkaloid interferes with the transport of the neuro-transmitter dopamine.
How many statements are correct ?

(1) 4

(2) 3

(3) 1

(4) 2

69) 

The above pyramid represents :-

(1) Standing crop

(2) Biotic potential

(3) Standing state

(4) Biomass

70) Apomixis is a form of :

(1) Sexual reproduction

(2) Asexual reproduction

(3) Syngamy

(4) Double fertilisation

71) Which of the following statement is correct about the rules of ICBN ?

(1) The first word represents the name of genus while second word represents the name of subspecies.

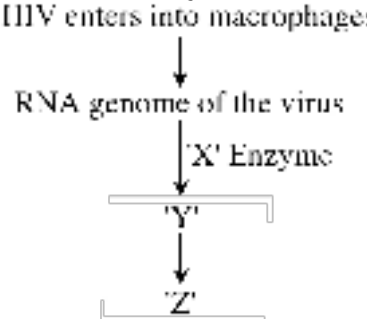
(2) Specific epithet starts with a capital letter.

(3) All letters of both generic name and specific epithet should be capital.

(4) Biological names are generally in Latin and written in italics.

72) After getting into the body of the person, the HIV enters into macrophages where RNA genome

of the virus replicates to form Y with the help of the 'X' enzyme. Now 'Y' gets incorporated into 'Z' and directs the infected cells to produce virus particles. The macrophages continue to produce virus and in this way acts like a HIV factory. The flow chart given below shows the replication of HIV



Which of the following correct regarding 'X', Y and Z.

(1) X = Protease enzyme; Y-Viral DNA;
Z = Host cells DNA

(2) X = Ligase enzyme; Y = Host cells DNA;
Z = Viral DNA

(3) X = Reverse transcriptase enzyme;
Y = Viral DNA; Z = Host cells DNA

(4) X = RNA polymerase enzyme;
Y = Viral DNA; Z = Host cells DNA

73) **Statement I** :- Nearly all of the essential nutrients, 70-80 per cent of electrolytes and water are reabsorbed by PCT.
Statement II :- ANF can cause vasodilation and thereby decrease the blood pressure to increase GFR.

(1) Both statement are correct.

(2) Only statement I is correct.

(3) Only statement II is correct.

(4) Both statement are incorrect.

74) Which of the following are part of epidermal tissue system?

(1) Cuticle, Cambium

(2) Trichomes, Cuticle

(3) Root hair, Mesophyll

(4) Trichomes, Mesophyll

75) **Assertion (A) :** According to Alexander von Humboldt, within a region, species richness. Increases with increasing area but only up to a certain limit.
Reason (R) : Species richness refers to the number of species present in a given area.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

76) Diplotene of meiosis-I is marked by :

(1) Formation of synaptonemal complex

(2) Terminalisation of chiasmata

(3) Dissolution of synaptonemal complex

(4) Crossing over take place

77) Read Column-I, II & III identify their correct match-

	Column-I		Column-II		Column-III
A.	Steroid hormone	P.	Epinephrine	i.	Thyroid gland
B.	Iodothyronine hormone	Q.	Thyroid hormone	ii.	Adrenal medulla
C.	Amino acid derivative hormone	R.	Cortisol	iii.	Hypothalamus
D.	Peptide hormone	S.	Oxytocin	iv.	Adrenal cortex

(1) (A–R–iv), (B–Q–i), (C–P–ii), (D–S–iii)

(2) (A–R–ii), (B–Q–i), (C–P–iv), (D–S–iii)

(3) (A–P–ii), (B–Q–i), (C–R–iv), (D–S–iii)

(4) (A–S–iii), (B–Q–i), (C–R–iv), (D–P–ii)

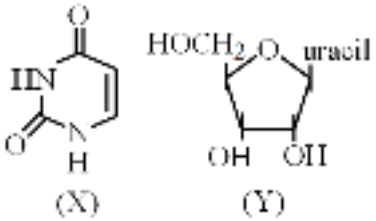
78) **Statement-I :** The testes are situated outside the abdominal cavity in a pouch called scrotum. The scrotum help in maintaining the low temperature of the testes (2-2.5°C lower than the normal body temperature) necessary for spermatogenesis. The testis is covered by dense covering
Statement-II : The male accessary glands include paired seminal vesicles, a prostate and paired bulbourethral glands. Secretion of these glands constitute the seminal plasma which is rich in fructose, calcium and certain enzymes.

(1) Both statement-I and statement-II are false.

(2) Both statement-I and statement-II are true.

(3) Statement-I is True and statement-II is false.

(4) Statement-I is false and statement-II is true.



(X) (Y)

79) What does X and Y represent in the structure given below ?

(1) X = Uridylic acid, Y = Uridine

(2) X = Uridine , Y = Uridylic acid

(3) X = Uridine , Y = Uricil

(4) X = Uracil, Y = Uridine

80) A flower is modified shoot, in which the apex produces different kinds of floral appendages laterally at successive nodes instead of _____.
Select the correct option :

(1) Leaves
(2) Buds
(3) Branches
(4) Fruits

81) Which of the following option include only Australian marsupials.

(1) Numbat, Tasmanian wolf, Bobcat
(2) Lemur, Flying squirrel, Flying phalanger
(3) Mole, Wolf, Mouse
(4) Bandicoot, Wombat, Koala

82) Detritivores break down detritus into smaller particles, this process is called :-

(1) Leaching
(2) Catabolism
(3) Humification
(4) Fragmentation

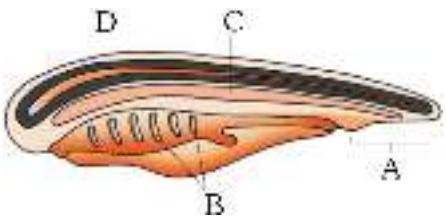
83) Toxin which is responsible for chill and high fever during malaria.

(1) Haemozoin
(2) Heam
(3) Haemoglobin
(4) Haematin

84) **Statement-I :** Golgi apparatus remains in the close association with the endoplasmic reticulum.
Statement-II : Materials to be packaged in the form of vesicles from the ER fuse with the cis face of the golgi apparatus.

(1) Only statement I is correct.
(2) Only statement II is correct.
(3) Both statements are correct.
(4) Both statements are incorrect.

85) In given fig identify A, B, C, D respectively



(1) Post-anal tail, gill slits, nerve cord, notochords

(2) Post-anal tail, gill slits, notochord, nerve cord

(3) Gill slits, post anal tail, notochord, nerve cord

(4) Gill slits, nerve cord, post anal tail, notochords

86) In the given examples, how many animals are viviparous ? *Corvus, Columba, Macropus, Pteropus, Calotes, Hemidactylus, Psittacula, Struthio* and *Macaca*

(1) Three

(2) Four

(3) Five

(4) Six

87) The correct sequence of steps in the experiment conducted by Hershey and chase is as follows;

(1) Centrifugation → Blending → Infection

(2) Infection → Centrifugation → Blending

(3) Blending → Infection → Centrifugation

(4) Infection → Blending → Centrifugation

88) The chemiosmotic hypothesis has been put forward to explain the mechanism of:

(1) ATP synthesis

(2) NADH synthesis

(3) Sugar synthesis

(4) Protein synthesis

89) Heterogamety is found in an organism with sex chromosomes-

(1) XY

(2) XO

(3) XX

(4) Both (1) and (2)

90) What are the possible blood types of the offspring of a cross between individuals that are type AB and type O ?

(1) AB or O

(2) A or B

(3) A, B or O

(4) A, B, AB or O

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	1	2	3	3	3	2	2	1	1	4	1	2	3	3	3	2	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	3	4	2	2	2	4	4	1	2	4	4	3	1	4	4	2	3	1
Q.	41	42	43	44	45															
A.	1	4	3	2	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	3	3	4	2	1	1	3	3	1	4	4	3	4	3	3	2	3	1	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	1	1	4	3	4	2	4	2	4	3	1	2	1	3	1	1	3	4	3
Q.	86	87	88	89	90															
A.	3	1	4	2	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	3	1	1	2	4	2	2	4	3	3	4	3	3	2	2	3	3	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	3	1	2	2	1	3	2	4	1	2	2	2	2	2	2	1	2	4	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	2	2	2	3	4	4	2	4	3	4	4	3	4	2	3	2	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	1	4	1	2	4	2	2	2	4	3	2	2	2	3	1	2	4	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	1	3	2	1	4	1	4	2										

SOLUTIONS

PHYSICS

1) $\frac{hc}{\lambda_3} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$

$\lambda_3 = \frac{\lambda_1\lambda_2}{\lambda_1 + \lambda_2}$

2) T.E. = $(PE)_{min}$ + $(K.E.)_{max}$

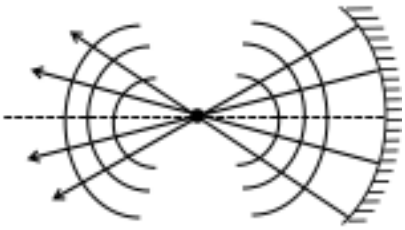
9 = 5 + $(K.E.)_{max}$

$(K.E.)_{max}$ = 4 J

This is KE at mean position

3)

After reflection from concave mirror



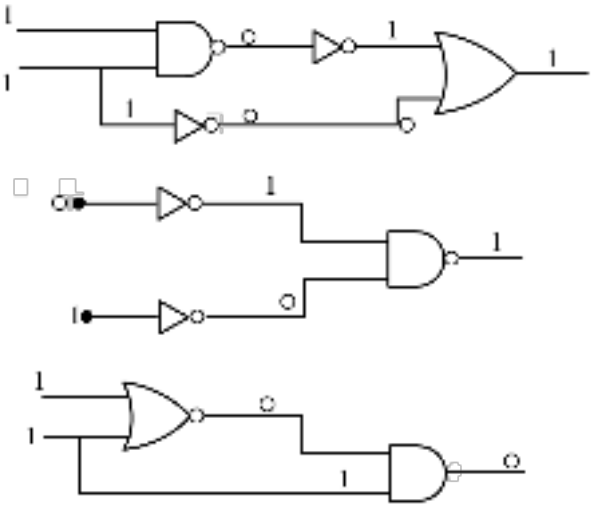
4) Heat lose = Heat gain

100 × s × (885 – 75)

= 100 × s(75 – 30) + 170 × 1 × (75 – 30)

s = 0.1 cal/gm°C

5)

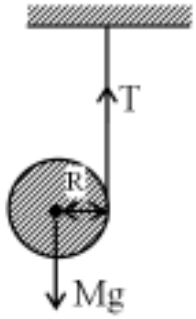


6) $V_{sound} = \sqrt{\frac{\gamma RT}{M_w}}$

$V_{sound} \propto \sqrt{\frac{\gamma}{M_w}}$

$\frac{V_{(N_2)}}{V_{(He)}} = \sqrt{\frac{\gamma_{(N_2)}}{\gamma_{(He)}} \times \frac{M_{w(He)}}{M_{w(N_2)}}} = \sqrt{\frac{(7/5)}{(5/3)} \times \left(\frac{4}{28}\right)}$

$\frac{V_{(N_2)}}{V_{(He)}} = \sqrt{\frac{3}{25}} = \frac{\sqrt{3}}{5}$



7)

⇒ Mg – T = Ma ...(1)

⇒ $T(R) = \left(\frac{1}{2}MR^2\right)\alpha$

$T(R) = \left(\frac{1}{2}MR^2\right)\left(\frac{a}{R}\right)$ [a = αR]

$T = \frac{1}{2}$ Ma put in equation (1)

Mg – $\frac{1}{2}$ Ma = Ma

⇒ a = $\frac{2g}{3}$

8)

According to Wein's displacement law,

$\lambda_{max}T = b$, where b is the Wein's constant

$T = \frac{b}{\lambda_{max}}$

Substituting the given values, we get

$T = \frac{2.88 \times 10^{-3} mK}{480 \times 10^{-9} m}$ = 6000 K

From the first balance point, we get $R/S = \frac{40}{60}$

After S is connected in parallel with a resistance of 60Ω, the resistance across the gap changes from S to S_{eq},

where $\frac{R}{60S/60+S} = \frac{50}{50}$

9) ⇒ R = 20Ω and S = 30Ω

10) Applying momentum conservation:

(80) 1 + 60 (–2) + (80 + 60 + 100) v = 0

$\boxed{v = +\frac{1}{6} m/s}$

11) Here $\frac{\Delta P}{P} = \pm\left(3\frac{\Delta A}{A} + \frac{5}{2}\frac{\Delta B}{B}\right)$

12)

Moment of inertia of a thin ring (A) about its central axis: $I_A = MR^2$

Moment of inertia of a disc (B) about its diameter: $I_B = \frac{1}{4}MR^2$

Moment of inertia of a hollow cylinder (C) about its axis: $I_C = MR^2$

Moment of inertia of a solid cylinder (D) about its axis: $I_D = \frac{1}{2}MR^2$

Comparing results: $I_A = I_C$

$I_D = 2I_B = 2 \times \frac{1}{4}MR^2 = \frac{1}{2}MR^2$

Thus, $2I_B = I_D$

13) Statement-I is correct, Statement-II is correct, and

As $\frac{\omega}{k} = \frac{2\pi f}{\frac{2\pi}{\lambda}} = f\lambda = c$

Speed of em wave in vacuum is c for all waves, independent of their λ.

14) $I = I_0 \cos^2\theta$

I_0 : Intensity of plane polarised light

I : Intensity of transmitted light

θ : Angle between transmission axes of polariser and analyser

θ = 0°, I = I_0

θ = 90°, I = 0

15) $v_{i/m} = -\left(\frac{v}{u}\right)^2 v_{O/m}$

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=\frac{1}{60}-\frac{1}{40}\Rightarrow v=-120$$

$$\left|v_{i/m}\right|=\left|-\left(\frac{120}{60}\right)^2\cdot 10\right|=40\text{cm/sec}$$

16) We know that in the given condition $s\propto\frac{1}{m^2}$

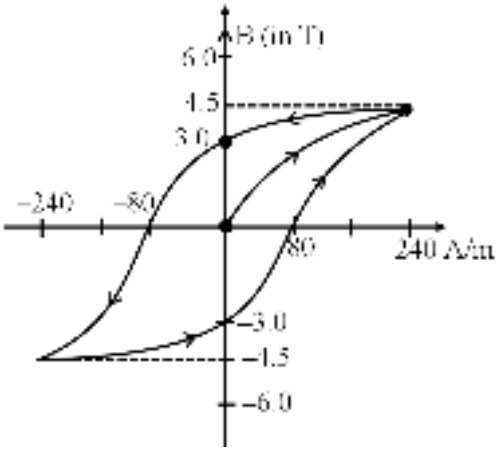
$$\Rightarrow\frac{s_2}{s_1}=\left(\frac{m_1}{m_2}\right)^2$$

$$\Rightarrow s_2=\left(\frac{m_1}{m_2}\right)^2\times s_1$$

17) $[h]=[M^1L^2T^{-1}]$, $[G]=[M^{-1}L^3T^{-2}]$,
$[U]=[M^1L^2T^{-2}]$, $[L]=[M^1L^2T^{-1}]$

$$\Rightarrow\left[\frac{mh^2G}{U\,L}\right]=\frac{[M]\left[M^1L^2T^{-1}\right]^2\left[M^{-1}L^3T^{-2}\right]}{\left[M^1L^2T^{-2}\right]\left[ML^2T^{-1}\right]\left[L^2\right]}$$

$=[LT^{-1}]$

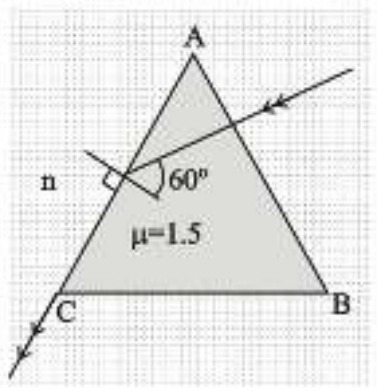


18)
Retentivity = 3.0 T
Co-ercivity = 80 A/m
Saturation = 4.5 T

19) Acceleration $=\frac{m_2}{m_1+m_2}\times g=\frac{1}{2+1}\times 9.8=3.27\text{m/s}^2$
and $T=m_1a=2\times 3.27=6.54$ N

20) For parallel coils

$$\frac{1}{t_{eff.}}=\frac{1}{t_1}+\frac{1}{t_2}\Rightarrow t_{eff}=\frac{10\times 15}{10+15}=6\text{ mins.}$$

21) 

Using snell's law at face AC

$1.5 \sin 60° = n \times \sin 90$

$1.5 \times \frac{\sqrt{3}}{2} = n = \frac{\sqrt{x}}{4}$

$3\sqrt{3} = \sqrt{x}$

$x = 27$

22) Distance covered by both the cars during this interval of time is same. That is

$$x = \frac{(v_1 + u_1)t}{2} = \frac{(v_2 + u_2)t}{2}$$

or $v_1 + u_1 = v_2 + u_2$

or $25 + 10 = v_2 + 15$

This given $v_2 = 20\ ms^{-1}$

23) Using energy conservation:

$KE_i + PE_i = KE_f + PE_f$

$\vec{P_1} = P\hat{i}$ $\qquad$ $\vec{P_2} = -P\hat{i}$

$$O + \frac{2KP}{a^3} \times P = \frac{1}{2}mv^2 \times 2 + 0$$

$$V = \sqrt{\frac{2P^2}{4\pi\varepsilon_0 a^3 m}} = \frac{P}{a}\sqrt{\frac{1}{2\pi\varepsilon_0 am}}$$

24)

$$I = \frac{10-4}{1+2+3} = 1A$$

from A to B

25) $A_1V_1 = A_2\ V_2$

$750 \times 10^{-4}\ V_1 = 500 \times 10^{-6} \times 0.3$

$$V_1 = \frac{500 \times 3 \times 10^{-3}}{750} m/s$$

$= 2 \times 10^{-3}$ m/s

if h is height of water in tank than $v_1 = -\frac{dh}{dt}$

$\frac{dh}{dt} = -2 \times 10^{-3} m/s$

26)

• Energy Distribution in Decay: The kinetic energy of the α particle is a fraction of the Q-value, depending on the mass numbers.
• Q-value: The total energy released in a nuclear reaction, shared between the products as kinetic energy.
The kinetic energy of the alpha particle (K_α) in an α-decay is related to the total energy released (Q-value, Q) by the mass numbers of the daughter nucleus (A_d) and the alpha particle ($A_\alpha = 4$). The formula used, derived from conservation of momentum, is:

$$K_\alpha = Q\left(\frac{A_d}{A_d + A_\alpha}\right)$$

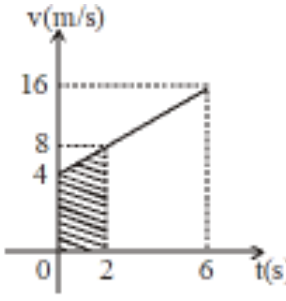
The mass number of the mother nucleus (A_m) is $A_d + A_\alpha$, so the formula can be rewritten in terms of A_m:

$$K_\alpha = Q\left(\frac{A_m - 4}{A_m}\right)$$

Given values are $K_\alpha = 98$ Me V and Q = 100 Me V. We substitute these into the formula and solve for A_m:

$$98 = 100\left(\frac{A_m - 4}{A_m}\right)$$
$$98A_m = 100(A_m - 4)$$
$$98A_m = 100A_m - 400$$
$$400 = 2A_m$$
$$A_m = \frac{400}{2}$$
$$A_m = 200$$

The mass number of the mother nucleus is 200.

27) 

$a = \frac{12}{6} = 2 m/sec^2$

Velocity at t = 2 sec.
v = u + at = 4 + 2 × 2 = 8 m/s
v = 8 m/sec

Disp. = Area under graph = $\frac{1}{2}[2][4 + 8] = 12$
$\Delta x = x_2 - x_1 = s \Rightarrow x_2 = s + x_1$
$x_2 = 12 - 4 = 8m$

28)

Poiseuille's equation

Rate of flow, $\frac{dV}{dt} = \frac{\pi P r^4}{8\eta \ell}$

$Q = \frac{dV}{dt} \propto Pr^4$

$\frac{Q_2}{Q_1} = \left(\frac{P_2}{P_1}\right)\left(\frac{r_2}{r_1}\right)^4 = 2 \times \left(\frac{1}{2}\right)^4 = \frac{1}{8}$

$Q_2 = \frac{Q}{8}$

29) For simple microscope,

$m = 1 + \frac{D}{f_0} \Rightarrow 6 = 1 + \frac{25}{f_0} \Rightarrow f^0 = 5$ cm

For compound microscope,

$m = \frac{\ell \cdot D}{f_0 \cdot f_e} \Rightarrow 12 = \frac{60 \times 25}{5 \cdot f_e} \Rightarrow f_e = 25$ cm

30)

$\therefore L = \frac{BNA}{l} = \frac{\mu_0 Ni}{2R} \times \frac{A}{l}$

$L = \frac{\mu_0 N \pi R^2}{2R} \Rightarrow L \propto R$

$\therefore L \propto \sqrt{A}$

31) $\vec{\tau} = \vec{M} \times \vec{B} \Rightarrow \vec{\tau} = 50\hat{i} \times (0.5\hat{i} + 3\hat{j})$

$= 150(\hat{i} \times \hat{j}) = 150\hat{k}\,N \times m$

32)

Coefficient of cubical expansion is

$\gamma = \alpha_x + \alpha_y + \alpha_z = \alpha_x + 2\alpha_y$ ($\because \alpha_y = \alpha_z$)

$= 2.0 \times 10^{-6} + 2 \times 1.6 \times 10^{-6} = 5.2 \times 10^{-6}\ K^{-1}$

33) In case (1) $C_{eq} = \frac{\in_0 \left(\frac{A}{2}\right)}{d} + \frac{K \in_0 \left(\frac{A}{2}\right)}{d}$

In case (2) $C_{eq} = \frac{\frac{\in_0 A}{d-x}\frac{K \in_0 A}{x}}{\frac{\in_0 A}{d-x} + \frac{K \in_0 A}{x}}$



Equating these two equation.

$x = \frac{2d}{3}$

so, $V_1 = \frac{A}{2}.d$

$V_2 = A.\frac{2d}{3}$

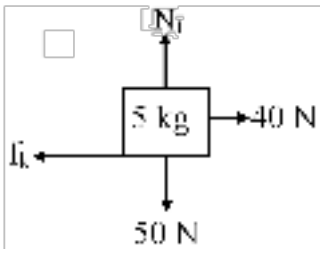
$\frac{V_2}{V_1} = \frac{4}{3}$

34) Since volume of water remains constant, then work done
$\Delta W = PdV = 0$
According to first law of thermodynamics
$dQ = dU + dW$, $dU = dQ = ms\Delta T$
$= 1 \times 4148 \times (70 - 40) = 4148 \times 30 = 124440$ J
$= 1.244 \times 10^5$ J

35) $F = \frac{-du}{dx} = 0$
$-(2x - 14) = 0$
$x = 7$

36) $V_i = (n+1)I_L R_S = V_L$
$R_S = \frac{V_i - V_L}{(n+1)I_L}$

37) $T = mg + m\omega^2 r = m\{g + 4\pi^2 n^2 r)$
$= m\left\{g + \left(4\pi^2\left(\frac{n}{60}\right)^2 r\right)\right\} = m\left\{g + \left(\frac{\pi^2 n^2 r}{900}\right)\right\}$

38) 

$N_1 = 50$ newton
$F_{\square} = \mu_{\square} N = 0.6 \times 50 = 30$ N
$F_k = \mu_k N = 0.5 \times 50 = 25$ N
$F_{app.} > F_{\square}$.
$a = \frac{f_{app} - f_k}{m}$
$a = \frac{40 - 25}{5} = 3 m/s^2$

39) $U = \frac{-GMm}{R}$

$$\Delta E = \frac{\Delta U}{2} = \frac{U_f - U_i}{2}$$

$$\Delta E = \frac{-GM_em}{8R_e} - \left(\frac{-GM_em}{4R_e}\right)$$

$$\Delta E = \frac{-GM_em}{8R_e} + \frac{GM_em}{4R_e}$$

$$\Delta E = \frac{GMm}{8R_e}$$

40) Acceleration produced due to electric field is

$$a = \frac{eE}{m} \qquad (F = ma = qE)$$

(u = 0) Now, $v = u + at = \frac{eEt}{m}$

de-Broglie wavelength of an electron is given by

$$\lambda = \frac{h}{mv} = \frac{h}{eEt}$$

Differentiate ⇒ $\frac{d\lambda}{dt} = \frac{-h}{eEt^2}$

41)

$\lambda_{micro} > \lambda_{IR} > \lambda_{uv} > \lambda_{gamma}$

42) $x = \sin\left(\frac{2\pi}{8}t\right) = \sin\left(\frac{\pi}{4}t\right)$

$$\alpha = -\omega^2 x = -\frac{\pi}{4} \times \frac{\pi}{4} \times \sin\left(\frac{\pi}{4} \times \frac{4}{3}\right)$$

$$= -\frac{\pi^2}{16} \times \frac{\sqrt{3}}{2} = \frac{-\sqrt{3}\pi^2}{32} \text{ cm/s}^2$$

43) Here,

$dW = PdV = \frac{RT}{V}dV$ (∵ PV = RT) ... (i)

Given $V = KT^{2/3}$

$$\Rightarrow dV = K\frac{2}{3}T^{-1/3}dT$$

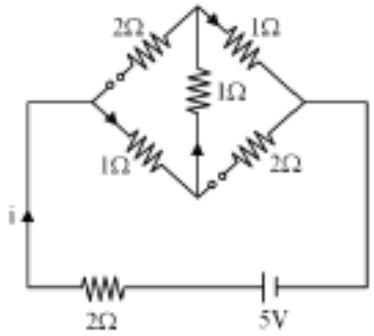
$$\Rightarrow \frac{dV}{V} = \frac{K\frac{2}{3}T^{-1/3}dT}{KT^{2/3}} = \frac{2}{3}\frac{dT}{T} \quad \text{... (ii)}$$

$$W = \int_{T_1}^{T_2} RT\frac{dV}{V} = \int_{T_1}^{T_2} RT\frac{2}{3}\frac{dT}{T}$$

From (i), [Using (ii)]

$$\Rightarrow W = \frac{2}{3}R(T_2 - T_1)$$

$$= \frac{2}{3}R \times 60 = 40R$$



44)

$R_{eq} = 5\Omega$

$i = \frac{5}{5} = 1$ Amp.

45)

$$E_1 = \frac{kQR}{\left(R^2 + R^2\right)^{\frac{3}{2}}} \;\&\; E_2 = \frac{kQ(2R)}{\left(R^2 + (2R)^2\right)^{\frac{3}{2}}};$$

$$\frac{E_1}{E_2} = \frac{5\sqrt{5}}{4\sqrt{2}}$$

CHEMISTRY

46)

$Na \xrightarrow{IE_1} Na^+ \xrightarrow{IE_2} Na^{2+}$

$3s^1 \qquad 2s^2 2p^6 \qquad 2s^2 2p^6$

due to change in shell no. $Z_{eff}\uparrow$

$\Rightarrow$ difference between IE_1 & IE_2 will be largest.

47) $P_A^o = 100$ mm $\qquad P_B^o = 80$ mm

$n_A = 1 \qquad n_B = 3$

$$P_{Th} = P_A^o X_A + P_B^o X_B$$

$$= 100 \times \frac{1}{4} + 80 \times \frac{3}{4}$$

$P_{TH} = 85$ mm

$P_{exp} > P_{Th}$

$\Rightarrow$ +ve deviation

(now options appears one by one)

48) Neutralisation between H_2SO_4 & KOH occurs

$\Rightarrow$ gram eq of H_2SO_4 = gram eq. of KOH

H_2SO_4	KOH
M = 2	W = x
V = 50 mL	vf = 1
vf = 2	

$$\frac{2 \times 50 \times 2}{1000} = \frac{x}{56} \times 1$$

$$x = \frac{56}{5} = 11.2 \text{ g}$$

$\therefore W_{KOH} = 11.2$ g

49) Benzoic acid Sodium Benzoate

M = 1 M = 2

V = x mL V = 300 – x mL

$pK_a = 4.2$ pH = 4.8

For acidic buffer solution

$$pH = pK_a + \log \frac{[salt]}{[Acid]}$$

$$\Rightarrow 4.8 = 4.2 + \log \frac{[2 \times (300 - x)]}{(1 \times x)}$$

$$\Rightarrow 0.6 = \log \left(\frac{600 - 2x}{x} \right)$$

$$\log 4 = \log \left(\frac{600 - 2x}{x} \right)$$

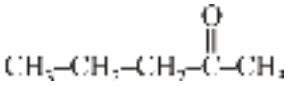
Taking antilog

$$4 = \frac{600 - 2x}{x}$$

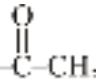
6x = 600

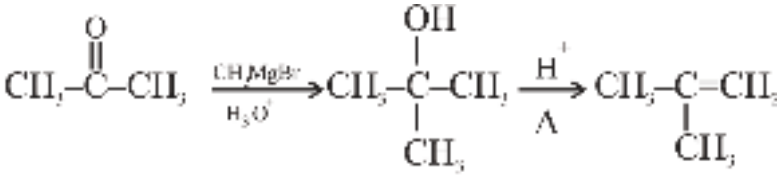
x = 100 mL

$V_{\text{Benzoic acid}} = 100$ mL

50) 

(2-Pentanone)

Aldehydes, Ketones having  group can give Iodoform test

51) 

Acetone P Q

52) NCERT-XI (Part-1), Pg. # 93

53)

PhCHO + HCHO $\xrightarrow{\text{Conc.NaOH}}$

$Ph–CH_2–OH + HCOO^{\ominus}Na^{\oplus}$
(Red.) (Oxi.)

54) $Ln + O_2 \rightarrow Ln_2O_3$
$Ln + H_2O \rightarrow Ln(OH)_3 + H_2$
$Ln + N_2 \rightarrow LnN$

55)

(P) Catalytic hydrogenation (H_2/Pd) of nitrobenzene reduces the nitro group ($-NO_2$) to an amino group ($-NH_2$). The product is aniline (C).

(Q) Catalytic dehydrogenation (Cu/Δ) of benzyl alcohol (a primary alcohol) oxidizes it to an aldehyde. The product is benzaldehyde (D).

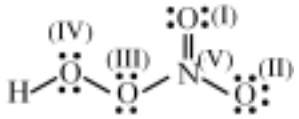
(R) Complete acidic hydrolysis (H_3O^+/Δ) of benzonitrile (-CN) produces a carboxylic acid (-COOH). The product is benzoic acid (A).

(S) Reduction of benzenediazonium chloride with H_3PO_2 removes the diazonium group and replaces it with hydrogen. The product is benzene (B).

Therefore, the correct match is P→C, Q→D, R→A, S→B.D)

56) Hydration Energy $\propto \frac{\text{Charge}}{\text{Size}}$

57) F.C. = (valence e^-) – $\left(\frac{\text{no. of covalent bond } e^-}{2}\right) - (\text{lone pair } e^-)$



I = 6 – 2 – 4 = 0
II = 6 – 1 – 6 = – 1
III = 6 – 2 – 4 = 0
IV = 6 – 2 – 4 = 0
V = 5 – 4 – 0 = +1

58)

$I^- + KMnO_4 \xrightarrow{H^+} \overset{+2}{Mn^{+2}} + \overset{0}{I_2}$

$I^- + KMnO_4 \xrightarrow{OH^-} \overset{+4}{MnO_2} + \overset{+5}{IO_3^-}$

x + y + p + q = 2 + 4 + 0 + 5 = 11

59) 1 mol $H_2O_{(\ell)} \rightarrow H_2O_{(s)}$

10°C – 10°C

Breaking down the process

ΔH_1 ΔH_2 ΔH_3

$H_2O_{(\square)} \rightarrow H_2O_{(\square)} \rightarrow H_2O_{(s)} \rightarrow H_2O_{(s)}$

10°C 0°C 0°C –10°C

$\Delta H_1 = + nCp \Delta T = 1 \times 75 \times (-10) = -750 \text{ J mol}^{-1}$

$\Delta H_2 = -\Delta H_{fusion} = -6 \text{ kJ mol}^{-1} = -6000 \text{ J mol}^{-1}$

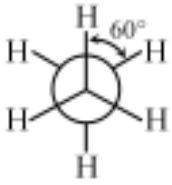
$\Delta H_3 = + nCp \Delta T = +1 \times 37.8 \times (-10)$

$= -378 \text{ J mol}^{-1}$

$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$

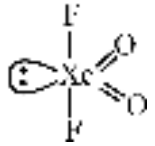
$\Delta H = -750 - 6000 - 378$

$\Delta H = -7128 \text{ J mol}^{-1}$

60) 

61) C.P. $\propto i \times C \propto \frac{1}{\text{F.pt.}}$

$Hg_2Cl_2 \rightarrow Hg_2^{2+} + 2Cl^-$

62) 

sp^3d, folded square

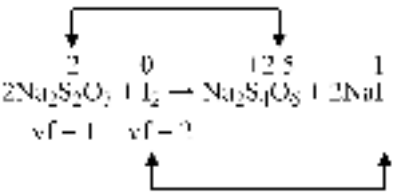
Bond angle : F – Xe – F > 0 = Xe = 0

Bond length : F – Xe > 0 = Xe

number of □p on Xe = 1



63) Iodine is reduced by hypo ($Na_2S_2O_3$)



g. eq of metal = g. eq of I_2 = g. eq of hypo.

$\frac{0.617}{E} = \frac{0.124 \times 46.3}{1000}$

E = 107.47

64)

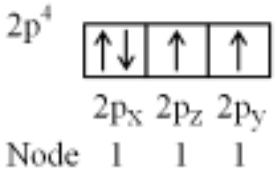
(1) Follow Huckel's rule $(4n + 2)\pi e^-$

(2) Planar sp^2
(3) Cyclic resonance

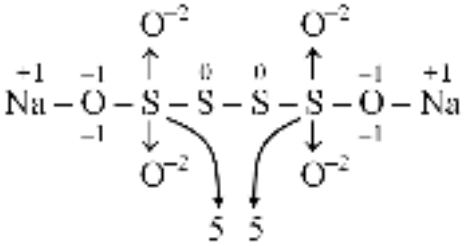
65)

(a) $Ag_3PO_4 \rightleftharpoons 3\,Ag^+ + PO_4^{3-}$
in acidic medium H_3PO_4 is formed.
Due to odd ion effect, equilibrium of Ag_3PO_4 shifts forward & solubility increases
It is correct
(b) Addition of Ar gas at constant pressure increases the volume
Hence, $P_{H_2O(g)} \downarrow \Rightarrow$ Reaction shifts forward
It is correct
(c)

$O \rightarrow 1s^2\ 2s^2\ 2p^4$
Zero node, 1 node, 3 node
Total node = 4



It is correct
(d) $Na_2\underline{S}_4O_6$



It is incorrect S → 5, 0, 0, 5

66)

$AB(g) \rightleftharpoons A(g) + B(g) \quad \left(\alpha = \frac{1}{3}\right)$

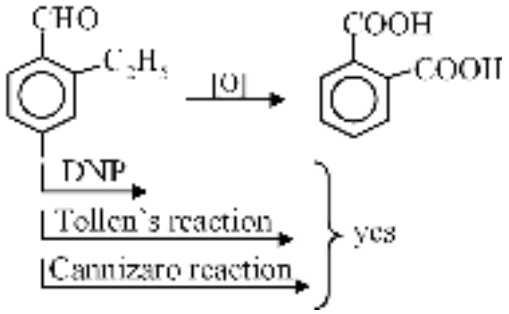
$1-\alpha$	α	α	
$\frac{2}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	
$P_{AB} = \frac{\frac{2}{3}}{\frac{4}{3}} \times P$	$P_A = \frac{\frac{1}{3}}{\frac{4}{3}}P$	$P_B = \frac{\frac{1}{3}}{\frac{4}{3}}P$	
$= \frac{P}{2}$	$\frac{P}{4}$	$\frac{P}{4}$	

$K_P = \frac{P_A \times P_B}{P_{AB}} = \frac{\frac{P}{4} \times \frac{P}{4}}{\frac{P}{2}} = \frac{P}{8}$
$\Rightarrow P = K_P \times 8 = 0.4 \times 8 = 3.2$ atm

67)

(1) Ma_2b_2 (sq. planar) → GI = 2, OI = 0
(2) Ma_2b_2 (Tetrahedral) → GI = 0, OI = 0
(3) Ma_2b_2 (Sq. planar) → GI = 2, OI = 0
(4) M $(AA)_2b_2$ (Octahedral) → GI = 2, OI = 2

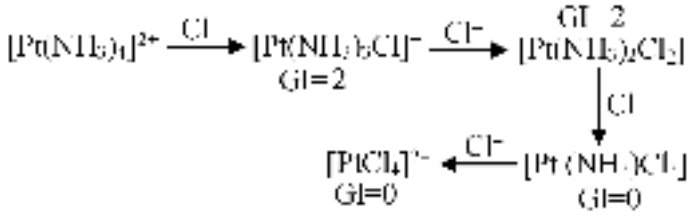
68)



69)

O - C_6H_5 has double bond character due to resonance.

70)



71) $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ $\Delta H = -2800$
6 mole 6 mole Heat rel. = 2800
now using unitary method

$\frac{6}{2800} \times 560$ $\frac{6}{2800} \times 560$ 560

$n_{CO_2} = 1.2$ $n_{H_2O} = 1.2$
$V_{CO_2} = 26.88\,L$ $W_{H_2O} = 21.6g$

72) Rxn-I 2Fe ..... $\Delta H_1^o = -800$ ...
Rxn-II $C_{(s)}$ ..... $\Delta H_2^o = -110$ ...
Rxn-III $3C_{(s)} + Fe_2O_{3(s)} \rightarrow 2Fe_{(s)} + 3CO_{(g)}$
ΔH_3^o...

III = II × 3 – I
$\Delta H_3^o = 3\Delta H_2^o - \Delta H_1^o$
$\Delta H_3^o = 3 \times (-110) - (-800)$
= – 330 + 800
$\Delta H_3^o = +470$
Now, $\Delta H^\circ = \Delta U^\circ + \Delta ng\, RT$

$$470 = \Delta U^\circ + \frac{3 \times \frac{25}{3} \times 300}{1000}$$

$\Delta U^\circ = 470 - 7.5$

$\Delta U^\circ = 462.5 \text{ kJ mol}^{-1}$

73)

$F_2 + H_2O \rightarrow HF + O_2$

$Cl_2 + H_2O \rightarrow HCl + HOCl$

$Br_2 + H_2O \rightarrow HBr + HOBr$

$I_2 + H_2O \xrightarrow{\Delta}$ [Non sponteneous Reaction]

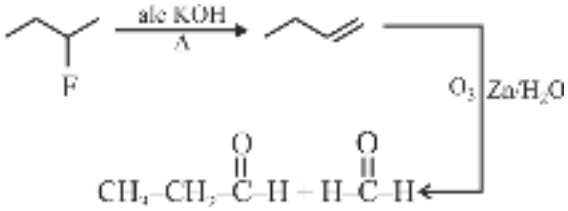
$F_2 + NaOH \rightarrow NaF + O_2$

$Cl_2 + NaOH \rightarrow NaCl + NaOCl$

$Br_2 + NaOH \rightarrow NaBr + NaOBr$

$I_2 + NaOH \rightarrow NaI + NaOI$

74)



75)

Species	Oxd. no. of underlined atom
$\underline{S}_8$	0
$\underline{Cl}O_3^-$	+5
$H_2\underline{O}_2$	–1
$\underline{P}_4$	0
$\underline{Cl}_2$	0
$\underline{Cu}^+$	+1
$\underline{N}O_2$	+4
K^+	+1
F_2	0
Na	0
Ag	0

76)

$BF_3 + F \rightarrow BF_4^-$

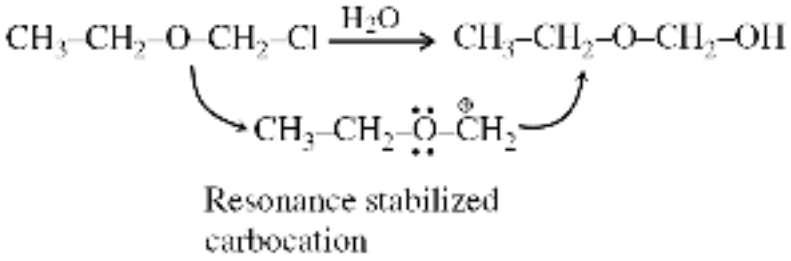
Non polar Non Polar

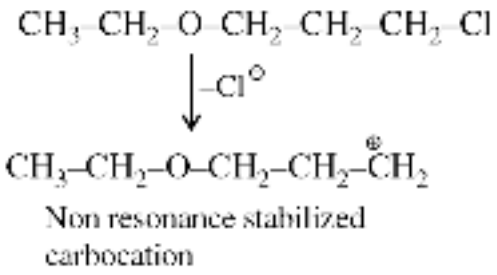
$PF_5 + F^- \rightarrow PF_6^-$
Non polar Non Polar

$ClF_3 + F^- \rightarrow ClF_4^-$
Polar Non Polar

$XeF_4 + F^- \rightarrow XeF_5^-$
Non polar Non Polar

77)





78) $K = 10^{-4}\ s^{-1}$ $E_a = 900\ kJ\ mol^{-1}$

$$\ell nk = \ell nA - \frac{E_a}{RT}$$

$$\ell n10^{-4} = \ell nA - \frac{900 \times 10^3}{8.314 \times 600}$$

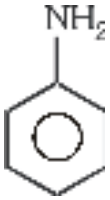
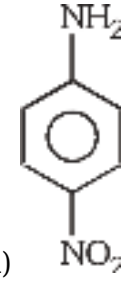
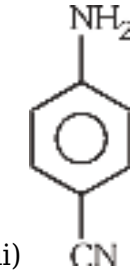
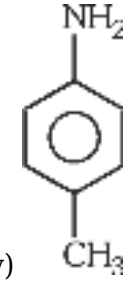
$-9.2 = \ell nA - 180.41$

$\ell nA = 171.21$

79) $H.O.H. \propto \frac{1}{\alpha H}$

80) **Explain the question :** Arrange the given compounds in decreasing order of basic strength.
Concept : This question is based on basic strength.
Solution : Basic strength $\propto \frac{+M, +HC, +I}{-M, -I}$

Given compounds	Remark

(i) 	This is the reference compound with no substituent.
(ii) 	–NO_2 is a strong EWG → decreases basicity
(iii) 	–CN is also an EWG, through less strong than NO_2.
(iv) 	–CH_3 is a weak EDG → increases basicity

Final order of basic strength (decreasing):

(iv) EDG (Strongest Base) > (i) No substituent (Reference) > (ii) Weaker EWG ($-NO_2$) > (iii) Strongest EWG ($-CN$) (Weakest Base)

Final Answer: 3

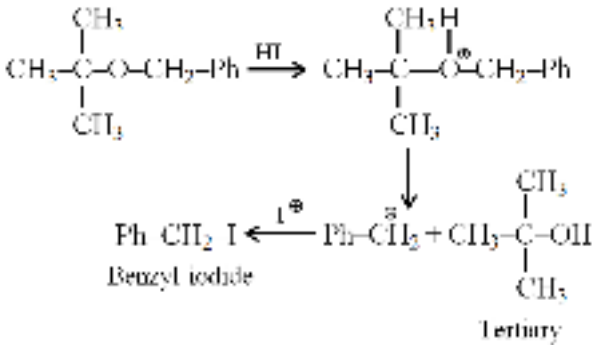
81) All sulphides are black except - ZnS, white , MnS, Pink, CdS - yellow

82)

RBC deficiency (pernicious anemia) occurs due to Vitamin B12 deficiency.

Other B-vitamins cause different diseases: B6 → convulsions, B1 → beri-beri, B2 → cheilosis.D)

83)



84)

$Cl^- + AgNO_3 \rightarrow AgCl \downarrow$ (white ppt)
$AgCl + NaCN \rightarrow Na[Ag(CN)_2]$ (Soluble).

85) $2H^+_{(aq)} + 2e^- \rightarrow H_{2(g)}$
Applying Nernst equation

$$E = \frac{-0.06}{2} \log \frac{P_{H_2}}{[H^+]^2}$$

$$0.06 = -\frac{0.06}{2} \log \frac{P_{H_2}}{(10^{-5})^2}$$

$$-2 = \log \frac{P_{H_2}}{10^{-10}}$$

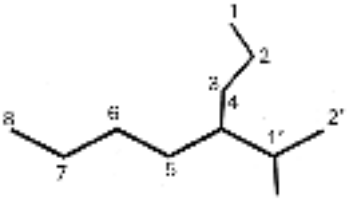
$$P_{H_2} = 10^{-12} \text{bar}$$

86) NCERT-XII (Part-I), Pg. # 122

87)

Explain from graph.

88) **Question Explanation:** IUPAC name of given compound is
Concept: IUPAC naming of hydrocarbon
Solution:



4-(1'-Methyl ethyl) Octane

89)

(A) Phenolpthalein is suitable indicator.
(B) For coloured solution upper meniscus is seen

(C) Benzenoid form of Hph is colourless
(D) Oxalic acid → Primary
$KMnO_4$ → Secondary

90)

Dipole moment of NF_3 is less than that of NH_3.
ClO_2 is odd e^- species
F.C. of O-atom in $SO_4^{2-} = -0.5$.

BIOLOGY

91)

NCERT, Pg. # 75

92)

NCERT XI, Pg # 114

93) NCERT-XI Page No. # 111

94)

Old NCERT-XI, Pg. # 112

95)

NCERT-XI, Pg. # 88, 98, 99

96)

NCERT-XI, Pg # 15

97) NCERT-XI, Pg # 171

98) NCERT-XI, Pg. # 43

99) NCERT Pg. # 292

100) NCERT Pg. # 186,187

101) NCERT-XI, Pg # 66

102) NCERT, Pg # 171

103) NCERT XI Pg # 173

104) NCERT Pg. # 243

105) NCERT Pg. # 227

106) NCERT Pg. # 121

107)

NCERT, Pg. # 72

108) NCERT Pg. 194-195

109) NCERT-XI, Page No. : 109

110) NCERT (XII) ; Page No. # 32

111) NCERT - XI, Pg.# 221

112) Module, Pg. # 9

113) NCERT-XII, Pg. # 169, 170

114)

NCERT-XI, Pg # 17, 18

115) NCERT Pg # 231

116) NCERT Pg. # 244

117)

NCERT-XI, Pg. # 45

118) NCERT-XII, Pg. # 112 & 116

119) NCERT Pg. # 132, 133, and 137

120) NCERT-XII, Pg. # 43

121) NCERT Pg. # 151 (E), 166 (H)

122) NCERT Pg. No. # 236

123) NCERT-XI, Pg. 87, 107

124) NCERT XIIth Page No. 199

125) NCERT-XI, Pg. # 13

126)

NCERT-XI, Pg. # 39

127) NCERT-XII, Pg. # 100,101,102 & 103

128) NCERT Pg. # 200

129) NCERT-XI, Pg. # 09

130) NCERT-XI, Pg. # 154

131) NCERT Pg.#205,207,208,210,213

132) NCERT-XI, Pg. # 101

133) NCERT Pg. # 189

134) NCERT Pg. # 105

135)

NCERT, Pg. # 173, 174

136)

NCERT-XI, Pg. # 63

137)

NCERT-XI, Pg. # 43

138) NCERT, Pg. # 169

139) NCERT XI, Pg # 15

140)

NCERT Pg. # 36 (E), 39 (H)

141) NCERT-XI, Pg.# 12

142)

NCERT-XI, Pg. # 137

143)

NCERT, Pg. # 225

144) NCERT, Pg. # 34, 35

145) NCERT-XI, Pg # 76

146) NCERT, Pg. # 61,62

147) NCERT-XI, Pg. # 83

148) NCERT XI Pg # 160

149) NCERT-XI Pg. # 177

150)

New NCERT, Pg. # 196

151)

NCERT, Pg. # 221

152) NCERT Class-XI, Page No. 121

153) NCERT XI, Pg. # 25

154) NCERT Pg., # 44,45 (E), 50,51 (H)

155) NCERT XI Page No.# 30-32

156) NCERT Pg#152

157) NCERT Pg # 247

158) NCERT-XII, Pg. # 158, 259

159)

NCERT, Pg. # 212

160) NCERT-XI, Pg. # 22

161) NCERT-XI, Pg. # 4, 5

162) NCERT, Pg. # 140

163) NCERT pg. # 209, 212

164) NCERT-XI, Pg. # 72-73

165) NCERT Pg. No. # 219, 220

166) NCERT XI_Page No.126

167) NCERT Pg # 248 (11th)

168) NCERT pg. no# 27,28

169)

NCERT, Pg. # 80,81

170) NCERT-XII, Pg. No. # 61

171) NCERT Pg. # 117, 118

172) NCERT Pg. No.#207.

173) NCERT Pg. # 131

174)

NCERT-XI, Pg. # 95, 96

175) NCERT Pg.# 55

176) NCERT-XI, Pg. # 50

177) NCERT, Pg. # 85,86

178) NCERT-XI, Pg.#140

179)

NCERT, Pg. # 70

180)

NCERT, Pg. # 61